Wolfson School of Mechanical, Electrical and Manufacturing Engineering

24WSP502META
Sports Engineering
MSc Individual Project Final Report

Boot-Surface Interaction on Artificial Turf for improving Mechanical Testing: understanding boot-surface interaction on artificial turf to refine testing standards and improve player safety

Taresh Gupta
Student ID F415206

Abstract:

Artificial turf has developed considerably since its introduction, with non-infill systems now promoted as a more sustainable alternative to traditional rubber-infilled 3G surfaces. Current certification standards, such as the Advanced Artificial Athlete (AAA) and Rotational Traction Athlete (RTA), are designed around infilled systems and may not fully capture the way players actually interact with non-infill carpets. This study set out as a pilot investigation to assess boot–surface interaction on non-infill football turf using in-vivo biomechanical testing, and to compare these outcomes with measurements from standard mechanical devices.

Five male football and rugby players performed repeated 180° shuttle-run pivots on three surface types: a new non-infill carpet, an artificially worn non-infill carpet, and a 3G sand–SBR infilled system. Motion capture (250 Hz) and force plate data (1000 Hz) were synchronised to calculate vertical and horizontal ground-reaction forces, rate of force development (RFD), coefficient of traction (CoT), and foot-segment behaviour. Complementary AAA and RTA tests were also carried out to record deformation, shock absorption, energy return, and peak torque.

The results showed that non-infill turf offered traction equal to, or slightly higher than, the infilled system, with CoT_plateau values of 0.6–0.85 and marginally higher horizontal forces (≈1.7 BW on fresh non-infill vs 1.65 BW on SBR), without an increase in vertical loading. Fresh non-infill produced faster and more consistent vertical loading rates (≈53 kN/s), whereas worn non-infill shifted towards higher combined peaks and steeper vertical RFD (up to 63 kN/s). In contrast, RTA

loads (<500 N) were far below the >2.5 × BW vertical and >2.0 × BW horizontal forces recorded in vivo.

Although constrained by sample size and some variability, this work provides initial player-based evidence for non-infill football turf. The findings point to a need for mechanical test thresholds that better reflect the demands of real player–surface interaction.

1 Contents

Tables

Figures

2 Introduction

2.1 Background

Artificial turf has undergone major technological changes since its introduction in the 1960s, progressing through three distinct generations. The latest third-generation (3G) systems, which became prominent in the late 1990s, addressed many shortcomings of earlier designs by incorporating longer fibres and performance infills. While (**Fédération Internationale de Football Association**) FIFA has approved their use for elite competition since 2004, adoption particularly in English football remains contested. Negative perceptions from second generation turf also poised a problem of carpet burns and skin abrasions. [1].

Modern advancements in sports surface technology have significantly altered the landscape of professional and amateur football. Over the past two decades, artificial turf has evolved from harsh, abrasive first-generation systems into highly engineered surfaces designed to mimic the performance and comfort of natural grass[2]. These surfaces are now deployed in top-tier competitions, training academies, and grassroots facilities alike. As part of this shift, there has been growing scrutiny surrounding ***boot–surface interaction***—the biomechanical interplay between a player's footwear and the turf—which plays a decisive role in both athletic performance and injury risk [3], [4] .

Figure 1 Evolution of football turf systems from natural grass to 4G synthetic turf[5]

Mechanical testing devices, such as the Advanced Artificial Athlete (AAA) and Rotational Resistance Athlete (RTA), are used as standard tools for certifying artificial pitches under FIFA Quality standards [5]. However, several studies have questioned the biofidelity of these devices. In particular, they have been shown to either overestimate or underestimate key parameters like rotational traction, energy return, and vertical deformation, largely due to their inability to replicate realistic human movement [6], [7]. For instance, there has been concerns over overestimating the window of parameters where real movement happens[8]. This mismatch creates uncertainty about the reliability of such tests in representing true injury risk or performance impact.

Figure 2 Boot Surface Interaction [10]

To address this gap, biomechanical research has increasingly shifted toward *in-vivo* methods that capture actual player movements using tools like high-speed Vicon motion capture and force plates. These systems allow for the calculation of critical variables such as peak ground reaction forces, coefficient of traction (CoT), and segment-specific foot dynamics[9], [10], [11]. This approach offers a more accurate reflection of real-world playing conditions, especially when evaluating complex football-specific tasks like cutting, pivoting, or shuttle runs [12].

The emergence of **non-infill artificial turfs**—engineered without loose rubber or sand—has added a new layer of complexity. These systems are designed to provide environmental and maintenance benefits by eliminating microplastics and reducing upkeep costs[13], [14] . Early commercial products have claimed promising mechanical performance; however, very limited biomechanical research has been conducted to validate these claims under real-world player movement [2], [15].

This thesis aims to evaluate how current mechanical testing standards compare against *in-vivo* motion capture data in measuring boot–surface interaction on **non-filled artificial turfs**. Specifically, it investigates whether current test methods underestimate or overestimate key biomechanical parameters when compared to player-derived data. By doing so, the study contributes to improving safety assessment, performance evaluation, and future turf certification frameworks.

2.2 Research Questions

This MSc Thesis is designed as a pilot study to investigate how studded football boots interact with non-filled artificial turf using in-vivo biomechanical testing. The study aims to address the following research questions:

1. **How does boot–surface interaction on non-filled artificial turf differ from existing data on 3G (infilled) surfaces** in terms of traction behaviour, torque generation, and surface deformation, GRF's experienced by the athletes?
2. Do current mechanical testing standards (e.g., **FIFA RTA and AAA**) overestimate or underestimate key biomechanical parameters—such as traction, torque, and deformation—on non-filled artificial turfs when compared to in-vivo player-derived data from Vicon motion capture and force plates?
3. How do **non-filled turfs** compare biomechanically to traditional **filled turfs** in terms of player–surface interaction, especially under high-traction movements?
4. **What boundary condition adjustments or algorithmic changes to current mechanical turf tests could improve their alignment** with real player data, especially on non-filled surfaces?
5. **What are the implications of these findings for surface designers, footwear manufacturers, and regulatory bodies** (e.g., FIFA) aiming to balance performance, safety, and environmental concerns in turf testing?

2.3 Project Aim and Objectives

Based on the research questions, the project defined the aim and objectives detailed in Table 1.

Project Aim
To quantify and characterize boot–surface interaction on non-filled artificial turf using in-vivo biomechanical data, and to evaluate how these findings can inform and improve mechanical testing standards for enhanced realism, safety, and performance relevance.
Project Objectives
Objective 1: “Review the existing literature on boot–surface interaction, traction mechanisms, and limitations of current mechanical test standards (e.g., FIFA RTA, AAA).” **Objective 2:** “Identify the key biomechanical parameters to be captured—such as ground reaction forces, coefficient of traction, torque ,peak forces—and design an appropriate in-vivo test protocol using Vicon motion capture and force plates.” **Objective 3:** “Capture and analyze in-vivo player-derived data during football-specific movements on non-filled artificial turf.” **Objective 4:** “Compare the measured biomechanical parameters with those estimated or assumed in current mechanical test procedures, and assess their accuracy in representing real player–surface interactions.” **Objective 5:** “Compare surface performance across non-filled and traditional filled artificial turf systems to identify key differences in rotational traction, deformation, and slippage.” **Objective 6:** “Formulate recommendations for refining mechanical testing methods or thresholds based on biomechanical evidence, with the aim to support safer and more biofidelic artificial turf evaluation.”

Table 1 - Project Aim & Objectives

2.4 Project Approach

The project was structured into four work packages. It began with a literature review to identify key biomechanical variables relevant to boot–surface interaction, followed by the definition of experimental methodology and setup for in-vivo testing on non-filled turf.

Biomechanical data were collected using Vicon motion capture and force plates during football-specific movements like shuttle runs and cutting. This enabled accurate logging of forces, torques, and foot segment behaviours.

Next section focused on comparing this player-derived dataset with standard mechanical test outputs (e.g., FIFA RTA, AAA) to assess their realism. Findings were used to suggest improvements in mechanical testing protocols for non-filled artificial turfs.

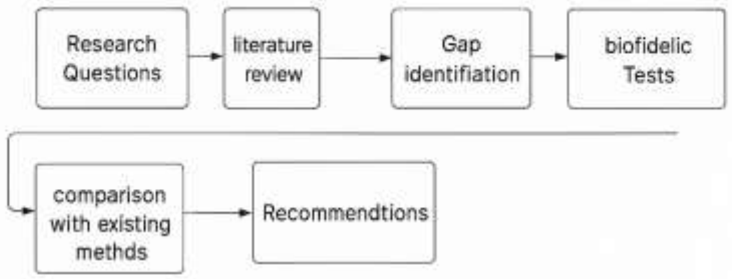


Figure 3.Research Strategy

2.5 Report Organisation

This report is organised into the following sections:

Section 1

Section 1 presents the project's introduction and literature review; the section is broken down into multiple subsections over two parts. Firstly, the context for the project is established by discussing the significance of boot–surface interaction in football and its relation to both performance and injury risk.

Part One

A detailed literature review introduces key biomechanical concepts related to boot–surface interaction, highlighting commonly measured parameters and methods of quantification. The limitations of current mechanical testing protocols, such as those outlined by FIFA (e.g., AAA and RTA), are discussed, alongside recent advances in marker-based and marker less motion capture technologies.

Part Two

The evolution of artificial turf technology is reviewed, with a focus on the development of non-infill systems. The section identifies a research gap in the biomechanical evaluation of non-infill turfs and sets the foundation for this project's experimental approach.

Section 2

This section outlines the experimental methodology, including the participant setup, motion capture and force plate configuration, trial design, and the parameters captured during the choice of movement on both filled and non-infill turf systems.

Section 3

Section 3 presents the data analysis and results. It includes comparative assessments of selected biomechanical variables such as traction, peak force and segment-wise foot behaviour between different turf conditions.

Section 4
This section interprets the findings in the context of prior literature. The discussion considers how the mechanical characteristics of non-infill turf may influence performance and safety, and whether current testing standards adequately represent real-world conditions.

Section 5
Section 5 summarises the study's conclusions with respect to the original research questions. Recommendations are proposed for improving mechanical test standards to better reflect in-vivo biomechanical realities.

3 Section 1: Literature Review

This Literature Review is divided into two parts. **Part One** explores the explanations of common keywords, key biomechanical concepts underpinning boot–surface interaction in football, including commonly measured parameters and how they are quantified. It critically examines the limitations of mechanical testing protocols from research papers—such as those defined in the **FIFA Quality Programme**[5], [16]—and evaluates advancements in biomechanical testing, including marker-based and marker less motion capture technologies.
Part Two reviews the evolution of artificial turf systems, with a particular emphasis on the development and performance claims of modern non-infill surfaces. It identifies a clear gap in independent biomechanical research on these surfaces and sets the foundation for the experimental investigation presented in later sections.

Part One: Biomechanical Concepts and Testing

3.1 Introduction to Boot–Surface Interaction

Boot–surface interaction refers to the complex biomechanical interplay between a player's footwear and the playing surface during both static and dynamic sporting movements [3]. This interaction is typically studied through movement patterns such as acceleration, deceleration, rapid directional changes, cutting, and stopping actions [9], or by analysing traction mechanisms[3]. When the studs of a football boot engage with the playing surface, they create frictional forces that oppose motion. These forces allow the athlete to push off effectively, change direction rapidly, or come to a stop. As such, boot–surface interaction is a critical link between athletic movement and surface properties, directly influencing performance and injury risk.

Dynamic movements commonly investigated in previous studies include acceleration from a stationary state, rapid deceleration, 130-degree cutting actions, pivoting, and sudden 180-degree turns [9]. Because shuttle runs combine multiple such dynamic actions, they are often used to evaluate parameters related to boot–surface interaction [10], [12].

Understanding boot–surface interaction is crucial from three complementary perspectives: enhancing athletic performance, preventing injury, and managing artificial turf systems. From a performance standpoint, optimal interaction ensures sufficient traction, enabling athletes to accelerate quickly, execute precise directional changes, and maintain postural stability with minimal slippage. Suboptimal traction—whether excessive or insufficient—can impair agility, increase the risk of foot entrapment, or negatively affect match performance[3], [6].

Injury rates in sports like football that use studded footwear are notably high, with approximately 22 injuries per 1,000 match hours and 3.5 injuries per 1,000 training hours reported [3]. The risk of injury increases significantly under poor surface conditions. Foot entrapment, caused by excessive traction, can fix the athlete's foot within the turf, increasing rotational torque on the ankle and knee joints. Such conditions have been linked to serious injuries such as anterior cruciate ligament (ACL) tears and other musculoskeletal trauma. On the other hand, insufficient traction may lead to slipping and instability, also increasing injury risk.

Beyond performance and safety, the surface itself—particularly artificial turf—plays a significant role. Despite its widespread adoption, artificial turf continues to raise concerns regarding its effect on lower extremity injuries. A systematic review of over fifty studies comparing injury rates on artificial turf and natural grass revealed no major difference in overall injury incidence [4]. However, a closer look showed a consistent pattern: foot and ankle injuries occurred more frequently on **artificial turf** than on natural grass, with knee injuries being the second most common [4]. These findings suggest that while the total injury burden may be similar across surface types, artificial turf may alter biomechanical loading or movement patterns, contributing disproportionately to injuries in the lower limbs.

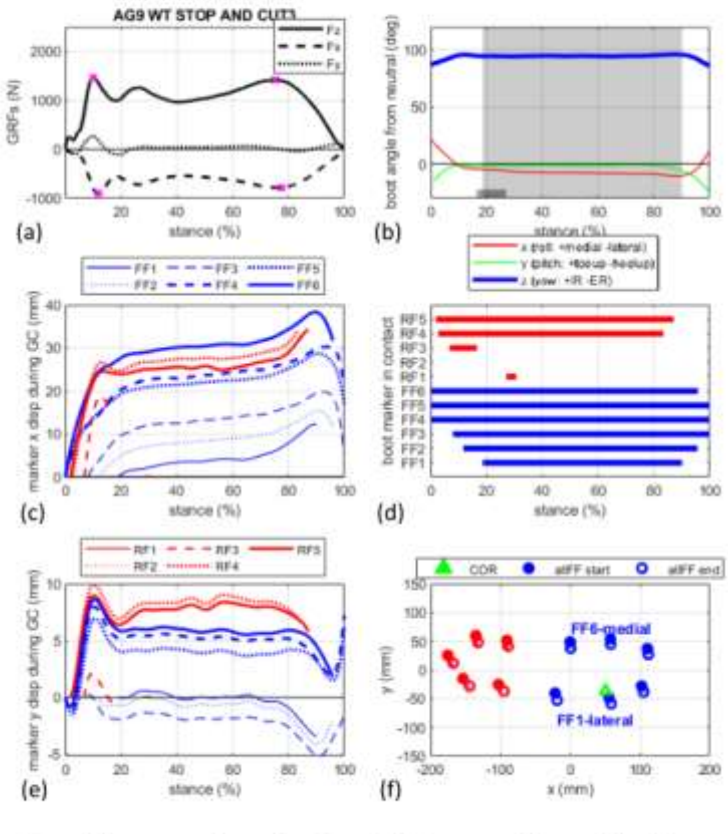


Figure 27 - summary figure of a stop and cut movement from an ideal player.

Figure 3 Boot Surface forces schematic investigated by [8]

3.2 Football Boot Outsole: Design and Traction Mechanisms

Football boot outsoles act as the critical interface between the athlete and the playing surface, influencing both performance and injury risk. These outsoles vary in stud shape, length, and layout, tailored for specific surface types—most commonly **Firm Ground (FG), Soft Ground (SG), Artificial Ground (AG)**. While each category is engineered for a particular surface, confusion often arises when FG boots are worn on synthetic pitches like 3G . FG boots, designed for hard natural grass, are frequently used on artificial turf due to convenience or habit. However, this crossover is not recommended, as their longer, blade-shaped studs can exert excessive pressure on the firmer synthetic surface, increasing the risk of injury. [17], [18].

In contrast, AG boots are specifically developed for synthetic surfaces, featuring shorter, hollow, plastic moulded studs that distribute pressure more evenly and reduce strain on the foot[18].

While studies [19] found that bladed FG outsoles generated significantly higher rotational torque than conical ones using mechanical testing, a player-based study observed no significant difference in the coefficient of traction (CoT) between these stud shapes during actual rotational movements[8] . Interestingly, AG boots with conical studs showed consistently higher CoT across all players, suggesting improved utilisation of turf traction. This contrast likely stems from methodological differences: the study used mechanical devices and different boot models with up to 40° rotation, while on another used consistent boot models and involved lower rotation. The findings reinforce that mechanical tests may not fully reflect real-world traction performance, as player behaviour can adapt to surface conditions [8].

Some mechanisms of traction were qualitatively described before[3]. The study proposed three primary mechanisms through which traction is generated during boot–surface interaction on 3G artificial turf.

1. **Mechanism 1**: The first is **outsole–surface shear resistance**, where the horizontal force applied by the outsole is resisted by the shear stiffness of the compressed infill and fibre layer beneath the boot.
2. **Mechanism 2: outsole–surface friction**, becomes dominant once the shear force exceeds the static friction threshold and sliding begins. This mechanism seems to be particularly important in **non-infill turf systems**, where there is no granular infill to deform or provide shear stiffness, making direct friction between the boot and the turf fibres the primary source of resistance. One especially interesting observation from Forrester and Fleming's analysis is that the surface in contact with the outsole is a complex mixture of fibres (free pile) and performance infill, meaning that the coefficient of friction (COF) lies somewhere between the values for each of these materials (estimated range: 0.2–1.084). Moreover, experimental evidence suggests that static and dynamic COF are very similar in these systems and can often be approximated using a single representative value.
3. **Mechanism 3: stud horizontal displacement resistance**, wherein each stud locally compresses and displaces the surrounding infill, forming a high-density zone that resists movement. As the stud continues to move, this resistance reaches a steady-state plateau once the growth and shear loss of the compressed infill zone reach equilibrium. These

mechanisms collectively highlight the complex, time-dependent nature of traction generation during real-world play, especially on synthetic surfaces.[3]

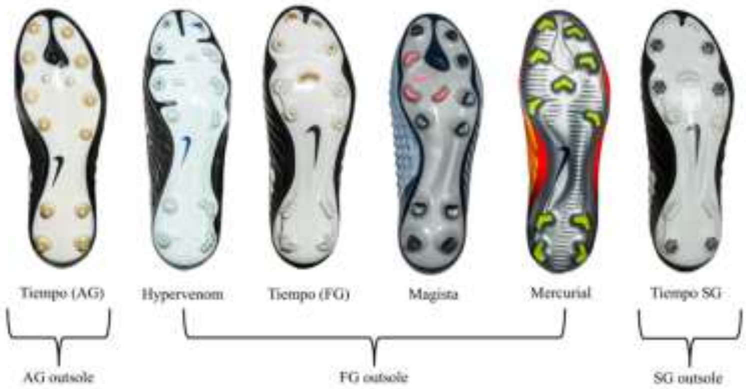


Figure 4Different types of boots identified by Thomson[17].

3.3 Mechanical Testing Standards (AAA, RTA)

FIFA's Quality Programme assures football turf performance and safety by standardising **player–surface** and **ball–surface** measurements in the lab and on installed fields. For player–surface behaviour, two devices dominate: the **Advanced Artificial Athlete (AAA)**, which characterises vertical impact response, and the **Rotational Traction athlete (RTA)**, which quantifies rotational resistance ("grip"). Surfaces must fall within defined Quality/**Quality Pro** limits using prescribed procedures, calculations, and reporting rules [5], [16].

3.3.1 Advanced Artificial Athlete (AAA)

What is AAA ?

The AAA is a controlled **spring–mass drop tester** designed to imitate the heel/foot impact seen in running. A **20 kg** instrumented mass—comprising an accelerometer, a linear spring, and a **70 mm** convex steel test foot—is held by an electromagnet and dropped vertically from a verified lift (nominally **55.00 ± 0.25 mm**) to ensure a narrowly bounded impact velocity. The mass is guided to maintain a single degree of freedom and to minimise parasitic friction; the accelerometer records the entire drop, impact, and rebound event for post-processing[5], [16] .

What does AAA calculates (general forms)?

Force–time from acceleration: Raw acceleration is expressed in multiples of g. Force is reconstructed as:

$$F(t) = m\,g\,[G(t) + 1];\ \ \boldsymbol{F_{max}} = \boldsymbol{m\,g}\,(\boldsymbol{G_{\mathbf{max}+1}})$$

where 'm' is the falling mass,
'g' is gravitational acceleration,

and G is measured acceleration in "g's"[5], [16] .

Shock Absorption (SA, %): SA reports the reduction in peak force relative to a concrete reference:

$$SA = (1 - F\_max\,/\,F\_ref) * 100$$

where F_ref = 6,760 N,
The shock absorption value **should be reported** to the nearest 0.1% (e.g., 56.9%) [1], [5].

Vertical (Peak) Deformation (mm):

Deformation is obtained by

(i) integrating the mass acceleration to displacement and
(ii) separating spring compression from specimen compression. During contact [T1,T2]

$$D\_specimen(t) = -D\_mass(t) + F(t) / C\_spring$$
$$Peak\ Deformation = max\ over\ t \in [T1, T2]\ of\ D\_specimen(t)$$

FIFA prescribes how to detect contact and how to report means across positions and environmental conditions[1], [5], [16] .

Energy Return (J). Energy returned to the mass is the **area under the unloading segment** of the force–deformation curve, computed with the trapezium rule; values are reported to 0.1 J.[5], [20].

How AAA results are generated and reported

FIFA's method ties calculation directly to verification and test conditions. The falling mass velocity and lift height are verified before testing; then, measurements are taken at specified positions and environments. For example, laboratory testing spans **23 ± 2 °C**, **–5 °C**, and **50 °C** conditions, with prescribed numbers of positions and drop sequences; field testing requires reporting of meteorological conditions. **Peak Deformation** (and the associated shock absorption (SA)/energy results derived from the same drop) are reported per location/condition with explicit averaging rules[5].

Earlier refinements to mechanical testing

A controlled 3G testbed study compared AAA outputs to **trained player perceptions** ("Leg Shock", "Give") across surfaces engineered to span FIFA ranges. **Vertical Deformation** consistently showed the strongest alignment with perceptions across most AAA configurations. Crucially, a **revised contact-detection and processing algorithm** which computes Energy Return from the **force vs deformation** curve rather than inbound/outbound velocities tightened agreement further; the study also observed **little difference** between using the **first drop** versus the mean of drops 2–3 for most configurations. These results support the AAA's construct validity *and* point to specific algorithmic updates to improve sensitivity without sacrificing reproducibility[20].

3.3.2 Rotational Traction athlete (RTA)

What is RTA?

The RTA measures the **rotational resistance** at the boot–turf interface using a circular **150 mm** test foot fitted with **six football studs**. The foot is normal-loaded (per the standard) and **rotated about a vertical axis** while torque (and, on instrumented versions, rotation angle) is recorded. FIFA recognises nominally equivalent devices: a heavyweight **RTT** and a **Lightweight RTT (LRTT)** and now also manually operated **RTA** operated at different **target rotational velocities**[1], [5], [20], [21].

What the RTA calculates (general forms)

(i) **Peak Torque (Nm).** The **primary FIFA metric** is the maximum of the torque–angle trace achieved during the prescribed rotation[5].

(ii) **Rotational Stiffness (Nm/°).** When angle is measured, a **stiffness** term or the slope of torque vs angle , can be derived within defined regions (e.g., a "secondary stiffness" between **50–80% of Peak Torque**), reflecting the **rate of traction build-up**. This variable captures how quickly resistance grows as studs plough and engage—an aspect closely linked to perceived slipperiness[20].

3.3.3 Rotational testing: speeds, player relevance, and reporting practice

FIFA sets different rotation speeds for the two traction devices: the heavy disc tester (RTT) is run at about 72 degrees per second with at least 45 degrees of turn, while the lightweight tester (LRTT) is run at about 30 degrees per second with 120 degrees of turn; a full circle at 72 degrees per second takes roughly five seconds, and at 30 degrees per second about twelve seconds [5], [16]. A multi-surface study that swept speeds from 10 to 210 degrees per second showed a clear rate effect: peak torque is lowest at very slow rotation, rises as speed increases, and on many infilled systems levels off once the test approaches the 72-degrees-per-second range; viscoelastic infills such as SBR, EPDM, and some cork blends showed the largest change with speed, whereas some organic infills and several non-infill carpets changed little [7]. Because the devices target different speeds, they also tend to report different values on the same spot: tests near 72 degrees per second read on average about 2–3 newton-metres higher than tests near 30 degrees per second. Three simple practices reduce this bias and improve comparability: use the same target speed when comparing devices, record the speed and angle achieved on every trial, and repeat any run that falls below the minimum intended speed [7].

In a controlled 3G setting, peak torque and a mid-range "secondary stiffness" derived from an angle-instrumented RTT aligned best with trained player perceptions of movement speed and slip, which supports keeping peak torque as the primary metric while adding stiffness to describe how traction builds under-foot [20]. Finally, rotational traction limits are set to balance slip risk against joint loading; top-tier certification accepts mean peak torque in the low thirties to mid-forties, while the general standard uses a wider band that stretches from the mid-twenties to about fifty newton-metres. Aligning rotation speeds across devices, logging achieved rate and angle, reporting a stiffness measure alongside the peak, and repeating below-threshold trials makes the numbers more comparable and closer to what players actually feel on the pitch [5], [16], [7], [20].

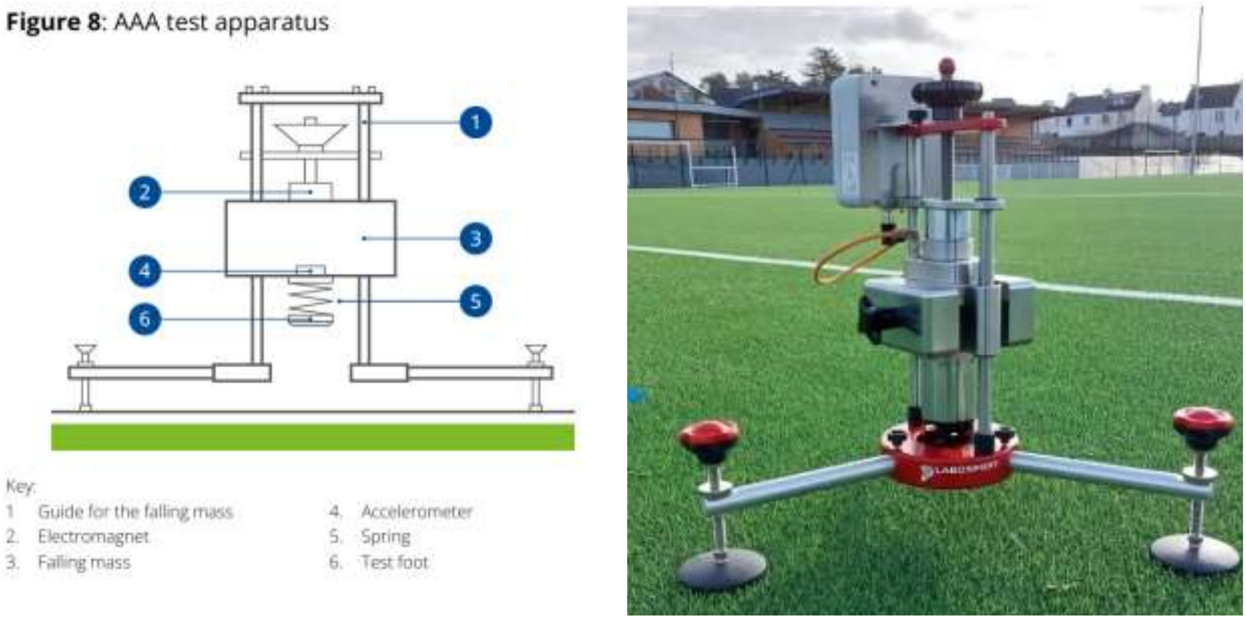


Figure 6. Advanced artificial athlete[5], [16], [22]

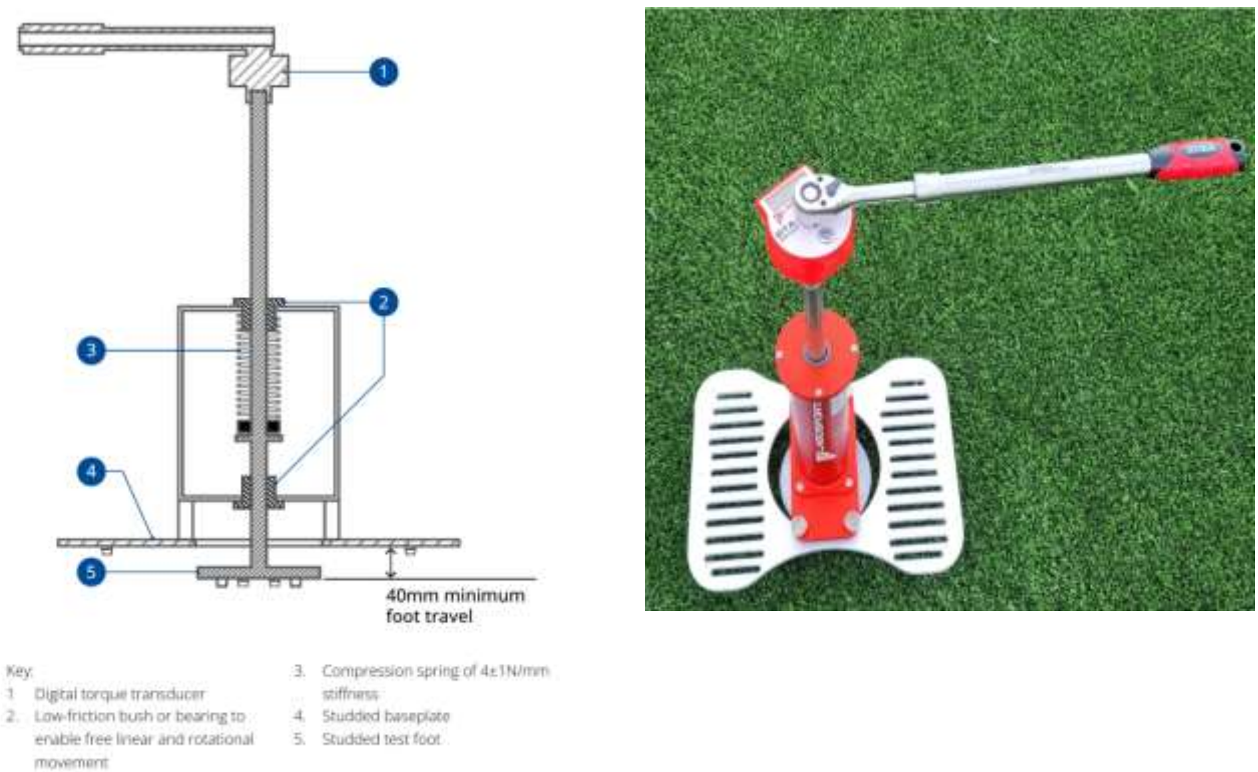


Figure 5. RTA [5], [16], [23]

Why these standards matter together

The AAA and RTA address **complementary** facets of the same interaction. The AAA yields physically interpretable **vertical response** metrics—**Shock Absorption** (force reduction vs concrete), **Peak Deformation** (how much the system compresses under load), and **Energy Return** (how much energy is given back)—using explicit, test-method-bound formulas and clear reporting rules (positions, environment, and averaging).

The RTA, meanwhile, quantifies the **rotational response** that governs *turning*, *cutting*, and *release* during stud–surface engagement. Standard practice focuses on **Peak Torque**, with growing support for **stiffness-type** variables that describe *how* traction builds. the push to log velocity and align device targets is gaining steady support[1], [7], [20], [23].

both devices now have **face-validity** against trained player perceptions in a controlled 3G setting i.e. AAA via updated signal processing for deformation/energy, and RTA via angle-instrumented stiffness, linking the **mechanical numbers** to what players actually feel under-foot. That is the core purpose of the standards are : **repeatable** measurements that meaningfully reflect on-field experience, while remaining practical for certification and monitoring without a complete biomechanical response study.

3.4 Parameters of Interest in Interaction

Accurately portraying how a studded football boot "negotiates" an artificial surface is a multi-scale problem. At one end sit **laboratory test rigs** that treat the surface as a passive material and impose controlled impacts or rotations; at the other end sit **biofidelic studies** that capture the emergent athlete–surface behaviour during real football manoeuvres. This section first defines the laboratory parameters embedded in the current **FIFA Quality Programme** and then juxtaposes them against the kinematic and kinetic variables measured in recent player-centred experiments. The goal is to show *where* the existing test envelope is robust and *where* it may need revision—especially for non-infill carpet systems, which now dominate small-sided facilities yet sit outside the historic infill-turf evidence base.

3.4.1 Laboratory Parameters and Devices in the FIFA Quality Manuals

Parameter family	Official device / test method	Core set-up (from FIFA 2024 manuals)	What the metric means for players	Current FIFA Quality range
Shock Absorption (%SA)	Advanced Artificial Athlete (AAA), FIFA TM 2024-03	20 kg guided mass drops from 55 cm onto a spring–foot system	% reduction in peak impact force vs concrete—cushioning of heel strike	60 – 75 % (Quality) / 60 – 70 % (Quality Pro)
Peak Deformation (VD, mm)	AAA, FIFA TM 2024-04	Same drop; vertical displacement of the mass at first impact	How far the surface compresses—linked to stability and energy cost	≤ 16 mm Quality / ≤ 15 mm Quality Pro
Energy Return (%ER)	AAA with accelerometer, FIFA TM 2024-05	Ratio of unloading to loading area under force–deformation curve	"Springiness"; may influence running economy	Informational only (no pass–fail)
Peak Torque (Tpk, Nm)	Rotational Traction Athlete (RTT), FIFA TM 2024-06	46 kg disc with six 12.5 mm plastic studs rotated to 90° at 12 ° s^{-1}	Worst-case rotational resistance during a planted turn—knee/ankle load	25 – 50 Nm Quality / 30 – 45 Nm Quality Pro
Rotational Shear Stiffness (RSS, Nm °$^{-1}$)	RTT, FIFA TM 2024-07	Slope of torque-angle curve between 50 % and 80 % Tpk	How abruptly torque rises—proxy for ACL loading rate	Informational only
Linear Friction / Translational Traction	Pendulum, slip-plate, or S2T2 rigs (non-standard)	Studded foot dragged at fixed speed & load	Grip for sprint & braking	No FIFA limit (research metric)
Ball–surface metrics	Vertical Ball Rebound TM 2024-01, Ball Roll TM 2024-02, etc.	Standard balls dropped or rolled on turf	"Game feel"; also, rough indicator of global surface stiffness	Multiple limits; e.g., Ball roll 4–15 m Quality

Table 2 Parameters checked in FIFA Quality Manual

3.4.2 Limitations Inherent to Laboratory Boundary Conditions

Under-loading – Real stop-cut foot strikes routinely exceed 2 kN. The 20 kg AAA hammer produces less than 1 kN, and the 46 kg RTT (not to be confused with RTA)platen produces similar compression loads. Player data [8] showed vertical GRF peaks of 1.4 to 1.6 kN that is even before accounting for elite sprinting speeds.

Over-rotation – The RTT twists the stud plate to 90 ° but match-representative cuts averages no more than 3–5-degree rotation [8].

Pivot mis-location – The RTT rotates the boot around its geometric centre, yet multiple optical studies locate the functional **centre of rotation (CoR)** under the lateral forefoot [11].

Operator dependency – Manual RTT devices still dominate field approval. [7] found that every 0.1 m increase in technician height reduced recorded Tpk(peak torque) by 0.1 Nm, enough to sway borderline passes for carpet systems that naturally yield low torques.[7]

Stud-geometry lock-in. The FIFA rotational-resistance rig is shod with **six round 12.6 mm × 18 mm PU studs** [5], [16], whereas contemporary AG soleplates rely on **"several small, short, round moulded studs"** rather than long blades or metal screws [17]. Because each of those AG studs is shallower and distributes load over a wider footprint, the lab disc sinks deeper and rotates about a larger torque arm than a real player's boot on low-profile turf—conditions that can **over-estimate peak torque while under-representing the true stud–surface contact area**.

These gaps are **particularly critical on non-infill turfs** and hybrid shock-pad systems because of absence of infill .

3.5 Limitations of Mechanical Testing

Mechanical tests are built for **repeatability and comparability**, not for reproducing the full biomechanics of play. That design choice creates a predictable **bio fidelity gap**: devices apply simplified loads, angles, and rates; constrain degrees of freedom; and often report peak-only metrics. **"It is not always possible to directly replicate a human interaction with a mechanical device due to limitations such as unrealistic normal loads and velocities**"[24]. Her in-vivo numbers illustrate the point—mean flat-foot rotation of **5.7°** (only occasionally more than > **20°**) and mean rotational velocity of **6.4°·s^{-1}**, which is well below the operating Rotational velocity widely used in device protocols [24].

(i) Kinematic mismatch (angles and rates). Standard rotational tests are built around a peak-centric event at relatively large angles. The FIFA method explicitly instructs operators to "Find the peak torque which represents the rotational resistance. Find the angle and the time corresponding to peak torque." [5]. This focus on a large-angle peak contrasts with human movement, where small-angle dynamics dominate early stance. Even on automated rigs, peak torque "commonly occurs between 30° and 40° of rotation," and at very high target speeds the device only reaches **95%** of that speed by **7.4°** of rotation—raising doubts about reproducing **small-angle** human motions[7]. Webb's in-play angles make the same point from the athlete side.

(ii) Peak rotational torque depends on how fast the device turns, especially at low speeds. When rotations were run from 10 up to 210 °/s across multiple fields, the lowest peak torques occurred at 10 °/s, and values rose with speed before flattening near the heavy-disc tester's target. On the same spot, tests at seventy-two degrees per second (RTT) read on average about 2.6 Nm higher than tests at thirty degrees per second (LRTT). The authors recommended using the same target speed on both devices, logging the speed actually achieved on every trial, and repeating any trial that falls below a minimum speed to avoid biased, low readings [7], [25].

Current standards already acknowledge this sensitivity by telling operators to repeat a trial if the mean rotational velocity falls outside a set window (for example, below twenty degrees per second or above fifty degrees per second on the lightweight device). However, this still anchors testing to a constant average rate rather than the short acceleration bursts seen in real play. It also leaves a loophole: an operator can start too slowly and then speed up to "hit" the average, even though much of the turn happened at a speed that under-reads torque. In practice, rate should be controlled and recorded, not just averaged, and any trial with prolonged low-speed segments should be discarded [5], [16].

(iii) Velocity effects are not the same on every surface. McGowan's tests showed that fields with viscoelastic performance infills—such as SBR, EPDM, or cork blends—produced higher peak torque as rotation speed increased, then levelled off near the heavy-disc target speed. In contrast, pine-woodchip infill and several non-infill carpets changed very little when the rotation speed was raised. This means that picking a single "standard speed" can favour one construction over another and bias cross-surface comparisons. The safer approach is to use the same target speed when comparing devices, record the speed actually achieved on every trial, and consider a short rate-sweep or normalisation by speed when you compare different constructions. In short, treat rotation speed as a controlled variable, not a background condition [7].

(iv) traction tests report only a single peak torque : single peak hides how resistance builds during the turn, because no angle data are stored. When the device is instrumented to record the full torque–angle trace, you can calculate stiffness and see the shape of the response. Cole and colleagues showed that a "secondary stiffness" taken from the mid-range of the curve (roughly half to four-fifths of peak) is a steadier indicator than a single maximum and can track construction changes more reliably (e.g., fibre state, infill condition) [1], [22].

Adding sensors also exposes how much operator speed can drift. In one dataset about six percent of trials had to be repeated because the achieved rotation rate fell below the lower limit of fifty-four degrees per second. This makes a strong case for always logging the actual rotation rate, setting clear pass/fail thresholds for each trial, and reporting both peak torque and stiffness metrics rather than relying on a single number [20].

(v) Algorithmic dependence in vertical tests (AAA): Device outputs depend on signal processing. Studies show Energy Restitution demonstrate the greatest benefit from the new algorithm from the force–deformation curve giving much stronger agreement with player attributes, and that Vertical Deformation increased in magnitude… increasing the range paired with a small reduction in test repeatability. Additionally, there was little difference between using the first drop or the mean drop[20]. This underscores a general limitation:

AAA "truth" is partly **algorithm-defined**, so method changes can shift values and rankings even when hardware is unchanged.

(vi) Loads, pressures, and geometry simplifications: For portability and control, devices use moderate, constant normal loads and simplified feet (e.g., **"six studded test foot"**) that do not capture outsole compliance or mixed stud geometries with the consequence being under-foot pressure during a simulated task was found to be **52 kPa** vs **26 kPa** in the RTT and around **15 kPa** in a translational rig at 70 kg.
Her field-portable rig also capped normal load at **46 ± 2 kg**, a common constraint in manual devices [24]. These are **method-class** issues.

(vii) Environment and reporting. Standards require reporting ambient conditions rather than controlling them in the field "Tests must be conducted under the meteorological conditions found at the time… The conditions must be reported."[5].
That improves transparency but leaves moisture/temperature/compaction as latent confounders unless studies move indoors specifically to stabilise surfaces.
In a study Ten 3G turf surfaces were constructed in an indoor testing area to ensure surface properties were controlled and not influenced by environmental conditions [1].

Bottom line. Current mechanical tests give **reliable benchmarks** but have inherent limits: peak-centric metrics at large angles, constant-rate actuation, sensitivity to rate (and to surface rheology), restricted sensing (without angle/penetration), algorithm-dependent outputs (AAA), simplified geometry and loads, and environment/operator effects. The literature already points to remedies: synchronise and **log** achieved velocity and **repeat** below-threshold trials, add angle sensing and report **stiffness**, adopt the improved AAA algorithm, and make environmental and surface-state reporting explicit . None of these eliminate the gap, but they move tests toward **biofidelity** while preserving the **repeatability** that certification needs.

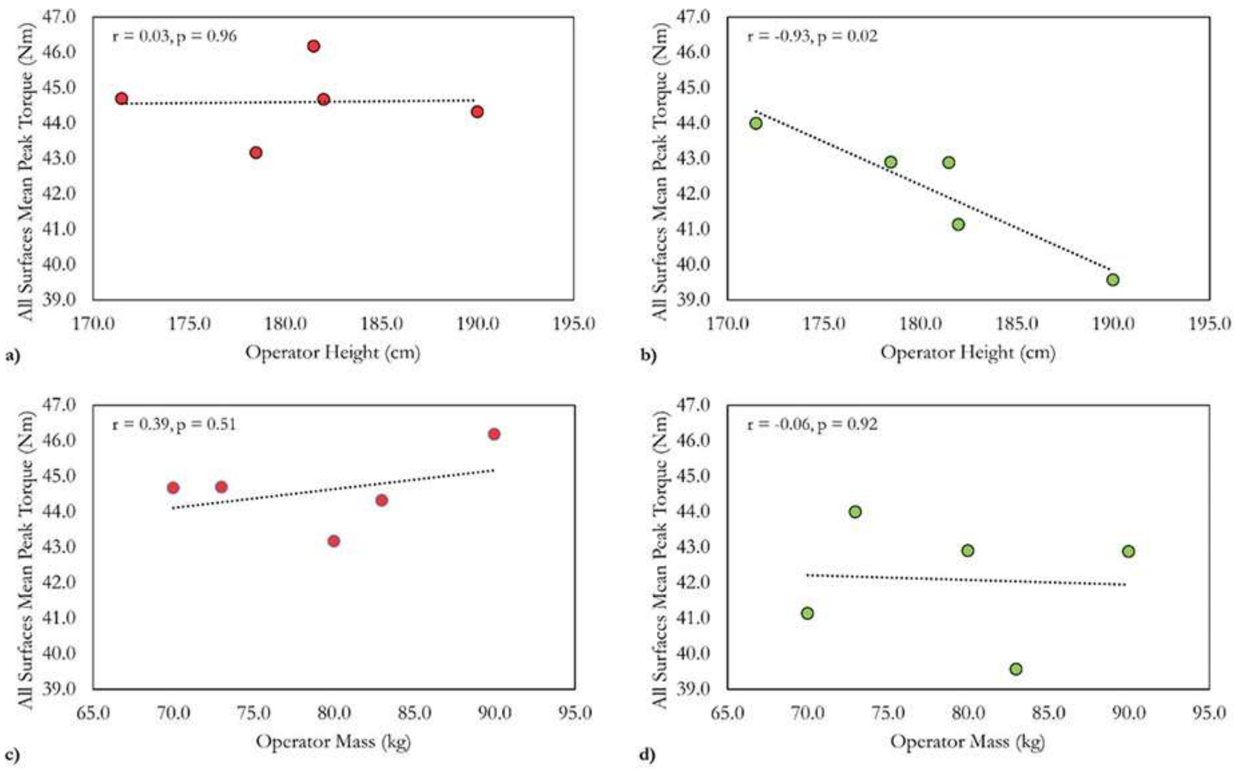


Figure 6 McGowan showed taller operators generated lower peak torque values using the LRTT.

3.6 Biomechanical Testing (Marker-based and Marker less)

This section brings together what studies with real players have shown on 3G turf i.e how hard players load the ground, how far they rotate while the foot is planted, how long contact lasts, how deep studs penetrate, and how much the condition of the surface matters—so we can use these realistic ranges later in the thesis[1], [24], [26].

Most of the player work sits in three streams. First, laboratory studies use marker-based motion capture and force plates. These give very accurate joint angles and both vertical and horizontal ground reaction forces. The trade-off is that the athlete has a short run-up and must land on a small force plate area, so the movement is more controlled but less “game-like”[1], [19], [27]. Second, on-pitch studies rely on high-speed video or marker less tracking. These let players move naturally and even slip for real, but the angles are noisier and the loads at the boot–surface interface must be inferred rather than measured directly [11], [28]. Third, a few studies add wearables such as pressure insoles or IMUs, which extend data collection into the field, but most published results still hinge on lab force plates or video analysis, meaning true shear and torque under the foot are rarely recorded [24], [26].

Contact time. Foot–ground contact in running, deceleration and cutting is short. Typical stance durations cluster around 0.10–0.30 s, and the first force rise can occur in the first 10–30 ms after touchdown[1]. In everyday terms, the key loading happens faster than a blink, so the early part of contact matters most.

Rotation while planted. When players cut or turn, the planted foot does not rotate very far while it is on the ground. Multiple studies report that most of the action sits within the first few degrees, commonly well under about 10–15° during stance; larger angles tend to happen right at touchdown or at toe-off rather than mid-stance[11], [26], [28]. This points to the importance of the initial “bite” rather than large, slow twists.

Loads. Vertical ground reaction forces scale with speed and the task being done. Walking is roughly one body weight; steady running sits around 1.5–3× body weight; hard decelerations and aggressive plant-and-cut actions on synthetic systems often reach about 2.5–3× body weight, with the first peak arriving very quickly [1], [19], [29]. For a player of 80 kg, that feels like briefly loading the leg with the weight of two to three people of the same size.

Penetration matters. High-speed video shows that most athletes land with either the heel or the forefoot first rather than landing flat. Because of that strike pattern, studs do not always go in as deeply as the outsole layout suggests, and shallower penetration generally reduces the traction the player can use[11], [28]. A simple way to picture it is pushing a fork into a bowl of rice: your angle and the state of the rice determine how deep it goes.

Surface dominates. Changes in surface condition—wet versus dry fibres, infill height and compaction—often shift performance more than moderate tweaks to the outsole. In timed slalom and cutting tests, wet turf and shorter studs slowed players, underlining that penetration and surface state drive how grip is mobilised [10], [12], [24].

Differences usually follow the measurement approach and the state of the pitch. Mechanical rigs can impose large angles and will often report high peak torques at those angles. Player tests, by contrast, show small stance-phase rotations and very fast loading, so the same surface can look “high-torque” on a rig yet behave as a “quick bite at small angles” under a player (Webb, 2017). Surface state flips results as well: the same boot can feel locked-in when the infill is dry and compacted but feel “skatey” when the rubber and fibres are wet, because both the centre of rotation under the foot and the achievable penetration depth change with moisture and compaction [10].

Most field and on-pitch studies do not measure shear force or torque directly at the boot–surface interface. Instead, they estimate loads from whole-body motion and force proxies, and they avoid truly injurious conditions for obvious ethical reasons [11], [26]. This means player studies are strongest when we use them to define realistic patterns and ranges rather than pass/fail thresholds.

Put simply, the player evidence tells us to value the **initial stiffness**—the slope in the first few degrees or millimetres—more than a single big peak torque at a large imposed angle. Mechanical tests should be set to player-realistic angles, contact times and loads, and torque should be presented alongside penetration and horizontal force so we can see how traction is actually mobilised on the pitch[24]. If a standard requires a peak number, we will still report it, but we will add early-phase metrics that match what players do to reach that stance [24], [26].

3.6.1 What Player-Based Studies Reveal

Study	Design	Key parameters	Boot vs Surface effect	Take-away
McGhie & Ettema 2013	Mechanical disk on 3G (50 mm infill)	Torque, traction coefficient	Bladed FG produced highest Tpk (≈ 40 Nm)	Strong stud-shape effect—but mechanical set-up rotated 40 °
Driscoll et al. 2015	In-game 3G, full-body mocap	Stud slip, CoR, yaw/pitch/roll	Slip driven by infill density; CoR lateral forefoot	Surface state more influential than outsole
De Clercq et al. 2014	10 × 5 m shuttles; dry vs wet turf; TF/AG/FG shoes	Traction ratio, shuttle time, slip count	Wet × TF (many small studs) ↑ slips (7/8)	Moisture amplifies surface effects, neutralises stud count
Newman 2023	Accel/decel & 130 ° cuts on fresh vs worn turf ; Puma shoes (AG vs FG)	CoT, peak forces, RFD, yaw, CoR	Only 3/48 outcomes significant – minor outsole influence	Modern players self-optimise; surface wear modest effect
Fleming 2011 review	Meta-analysis of 60+ mechanical vs player studies	—	—	Advocates merging player boundary conditions into FIFA tests

Table 3 Review of previous biomechanical studies

Common ideas among the major papers:

Player-based studies quantify what actually happens at the boot–surface interface during real football actions. They use motion capture and force plates to record how the foot loads and moves, then derive a set of variables that describe grip, stability and loading on the lower limb [2], [24]. The aim is to define the **boundary conditions** that matter for performance and injury and to check whether laboratory tests reflect these conditions [2], [24] .

Ground reaction forces (GRF). Vertical and horizontal GRFs show how hard the player loads the surface during running, braking and cutting. They are often normalised to body weight to allow comparison between athletes. Vertical GRF captures impact and push-off demands. Horizontal GRF reflects the traction needed to accelerate, decelerate and change direction[24], [30]. Player studies consistently report vertical GRF peaks around 1.5–2.5 BW in cutting tasks, with lower but important medio-lateral and antero-posterior components that drive slip/no-slip outcomes [24], [31], [32].

Contact time. This is the duration the foot stays in contact with the surface during a step. It frames how quickly forces develop and relax. Shorter contact times in sharp stops and turns mean loads and rotations must build quickly, which increases sensitivity to early-phase traction[24], [26].

Foot rotation (angle) and rotational velocity/acceleration. These describe how far, how fast and how abruptly the foot turns while planted. Multiple studies show that on-field rotations are **small** compared to laboratory test angles: the foot often rotates only a few degrees during the flat-foot phase, with most rotation clustered around touchdown and push-off [24], [26]. Rotational velocity and acceleration indicate how quickly torque rises; high initial rates can increase ligament loading if the boot "sticks" [2], [24].

Approach and impact velocity. Player speed at touchdown (both vertical and horizontal) affects stud penetration and the onset of traction. Higher approach speeds can increase initial penetration and the early horizontal resistance the surface must provide to prevent slip[24], [33].

Slip events. "Slip" is unintended boot motion relative to the surface. Counting slips and measuring their magnitude helps connect surface state (e.g., infill density or moisture) to performance loss and potential injury mechanisms[10], [11]. Fewer, smaller slips generally indicate more reliable grip in that task.

Centre of rotation (CoR): CoR is the point under the foot about which rotation occurs during a cut. In on-field measurements it is often located under the **lateral forefoot**, not at the geometric centre of the boot. This changes which studs engage and how torque builds and is a key reason to align lab tests with player kinematics[11].

Foot orientation in 3D (yaw, pitch, roll): Yaw (turning left–right), pitch (toes up–down) and roll (edge-tilt) describe how players "set" the foot to find grip while protecting the joints. Players often adjust pitch and roll to manage available traction and avoid excessive torsion[11], [24].

Coefficient of traction (CoT) / traction coefficient: This ratio of horizontal to vertical force is a simple way to express how much grip is used or available during a movement. Higher CoT supports faster acceleration and tighter cuts, but very high CoT can "fix" the foot and drive up joint loads [19], [34].It is different from **Traction ratio / utilised traction.** Some studies compare the **available** traction with the traction the player actually **uses** in the task. This shows how athletes self-moderate foot rotation and loading rather than exploiting the full mechanical limit of the surface [2], [10].Nomenclature of '**Traction**' is only used when the interaction involves studs ,spikes or cleats[24].It is also not the same as friction is to 'simplistic' to model the complex interaction that takes place in a boot surface interaction.

Performance metrics (e.g., shuttle time): Task times over short shuttle runs or cutting drills summarise whether a surface–boot combination supports effective acceleration, braking and redirection[10].

Peak forces and rate of force development (RFD): Peak vertical or horizontal forces indicate maximum loading demands. RFD shows how fast force rises after touchdown. High peaks and very rapid RFD can be performance-helpful but may increase tissue loading risk if traction is too high or too abrupt[2], [35].

Together, these parameters describe **what players actually do** to the surface and **what the surface gives back** during sport-specific actions. Consistent findings across studies include small in-stance rotation angles, short contact times, and strong sensitivity of slip and CoT to surface state (e.g., density and moisture) rather than to moderate changes in outsole. These observations define the biomechanical envelope that laboratory tests should aim to replicate, especially when interpreting rotational measures taken at large angles and constant speeds [2], [24].

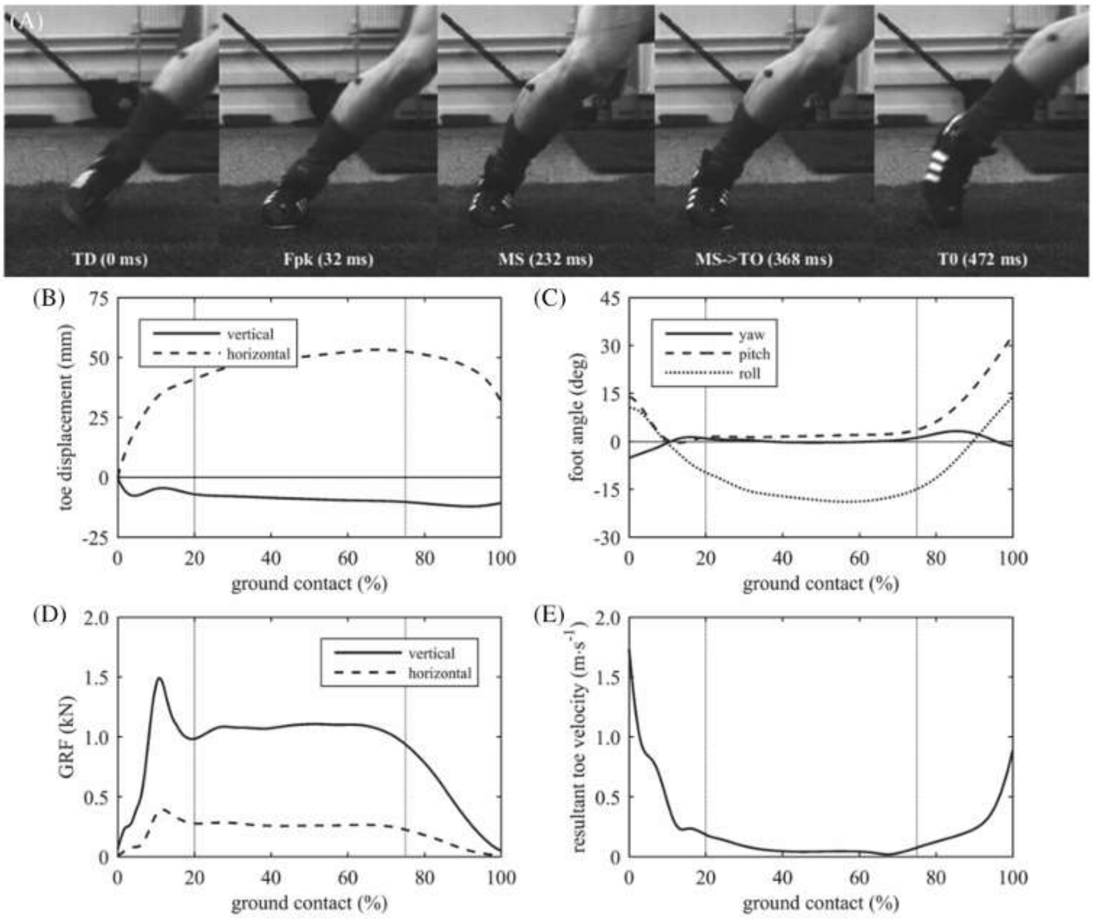


Figure 7 Typical data from the ground contact phase of a stop-and-turn movement on 3G turf. Key instances: touchdown (TD), loading peak vertical force (Fpk), midstance (MS), early toe-off (MS→TO), and toe-off (TO). Subplots: B) toe marker displacement, C) foot angles, D) ground reaction forces, and E) toe marker resultant velocity (Forrester)[3]

Part Two: Artificial Turf Systems

3.7 Evolution of Artificial Turf: From 1G to Non-Infill

3.7.1 What is artificial turf?

Artificial (synthetic) turf is a man-made sports surface that aims to reproduce the look and key play qualities of natural grass while offering high durability and all-weather availability. It does this with a textile “carpet” of green polymer fibres that stand in for grass blades, supported by sub-layers that carry loads, drain water, and tune how soft, springy, and grippy the pitch feels underfoot. The point is not to copy every aspect of grass, but to deliver safe, predictable player–surface interactions and consistent ball behaviour across heavy use and varied climates.

3.7.2 Construction and materials of artificial turfs

Although designs vary, most modern football systems are built from the same core parts:

- **Carpet and fibres.** Polyethylene fibres are tufted or woven into a backing to create the “pile”. Fibre type and geometry (monofilament vs. fibrillated, shape, texture, and tuft spacing) influence friction with the ball and how well the pile resists flattening. The backing is bonded so tufts don’t pull out .For example companies use a triple ‘W’ weave for safely bonding the fibres to the backing [36].
- **Stabilising infill.** A thin layer of rounded silica sand sits low in the pile to hold the carpet steady and keep fibres upright without contributing much to cushioning.
- **Performance infill.** A thicker, more compressible infill sits above the sand to create the player–surface interface. Historically this is SBR rubber crumb from recycled tyres; newer systems may use thermoplastic elastomer (TPE) or other engineered granules. Infill choice and condition affect grip and hardness, so it matters for passing accreditation tests and for long-term feel.
- **Shockpad (optional but common).** An elastomeric pad beneath the carpet provides extra cushioning and helps keep shock absorption stable over time. Pads may be prefabricated sheets/tiles or in-situ bound rubber.
- **Sub-base and drainage.** Bound layers and free-draining stone give a flat, durable platform and remove water quickly so matches can proceed in wet weather.

Traction and impact response come from the *combination* of fibre/infill interaction and the compliance of the pad below; changing one part shifts the whole system’s behaviour [2]

3.7.3 Generations of turfs

The “generation” labels are historical shorthand used by engineers and governing bodies to describe broad design shifts—*not* formal product names.

(i) **First generation (1G).** Dense, short nylon carpets with **no infill**. They were hard under impact and abrasive on skin; early football trials attracted criticism. Hockey adopted watered versions to reduce friction.

(ii) **Second generation (2G). Sand-filled** systems with slightly longer, softer fibres and wider tuft spacing to accept sand. These brought better stability and predictable ball roll, suiting hockey and multi-use community facilities. “Astro”-style dimple-soled footwear emerged for these pitches.

(iii) **Third generation (3G). Rubber-infilled long-pile** systems aimed squarely at football and rugby. Longer fibres stand above a deep layer of performance infill (rubber) over a stabilising sand bed; a shockpad is often included. The intent is to allow conventional studded boots and produce grass-like grip and cushioning. Modern FIFA/World Rugby approval frameworks are built around this class.

3.7.4 About "4G" and "non-infill"

In industry practice and standards, **there is no recognised '4G' category**. National and international bodies approve *systems* against performance tests, not "4G" labels. UK authorities and the grounds sector are explicit that "4G" is a **marketing term**, not an accredited generation. Some non-infill (dense, textured carpet + pad, no rubber granules) systems exist and may be marketed as "4G", but governing bodies focus on whether they *pass the tests*, not on that name[2].

The EU has adopted a REACH restriction on intentionally added microplastics that will phase out polymeric infills used on sports pitches after a long transition period, driving R&D into non-infill carpets and natural or encapsulated alternatives—but again, approval remains performance-based . Increased emphasis on sustainability in the artificial turf industry has shifted attention towards organic and non-filled surface systems; yet research surrounding new infill materials and surface systems is lacking. [37]

3.7.5 Non-filled turfs

Non-infill football turf replaces loose performance materials with an engineered carpet–pad system. The carpet is built dense, with a texturised "thatch" layer in the root zone and a high stitch rate to hold the pile upright, stabilise underfoot grip, and contribute to energy absorption. A separate shockpad typically carries more of the impact work so that the surface can meet targets for shock absorption and deformation without relying on loose particles. Performance is tuned through the carpet–pad design—yarn shape and texturization, pile height and density, backing stiffness, and the thickness and modulus of the pad—so that traction, ball roll and rebound sit within the same field test windows used for filled systems. Day to day, these fields install faster, run cleaner in use, and avoid end-of-life handling of loose materials; maintenance focuses on brushing to recover pile and monitoring pad condition rather than decompaction or top-ups. The design challenge is long-term stability: traction, resilience and ball control must be delivered consistently by the carpet–pad pairing as fibres age and traffic concentrates in known hot spots. "4G" is a marketing label rather than a recognised standards category, and non-infill fields are certified against the same FIFA/FA test suite for shock absorption, vertical deformation, traction and ball behaviour as infilled pitches [2], [36], [38], [39], [40].

In practice, non-infill systems have gained early adoption on small-sided and community pitches, with elite-level pilots now emerging as carpet–pad designs mature. Ball–surface control is achieved by combining higher pile densities with textured yarns that manage roll and rebound without auxiliary loose materials. Specifying an appropriate shockpad is critical for cold-weather hardness and for keeping force reduction within target ranges over the life of the field. These systems also help facilities respond to policy pressure on polymeric microplastics by eliminating intentional infill release pathways and simplifying end-of-life recycling as claimed by manufacturers. [13], [15], [36], [38].

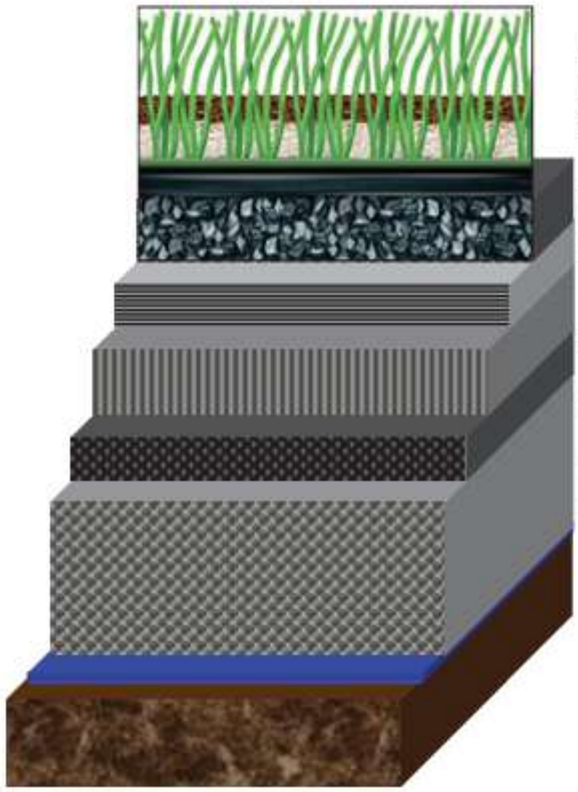


Figure 8 Components of a 3G turf

3.8 Gaps in Research on Non-Infill Turf

Most research on boot–surface interaction still relies on mechanical rigs and simplified loading, with only a small number of studies attempting biofidelic traction measurements on modern, infilled artificial turfs. The critical gap for this thesis is that comparable player-based or biofidelic evidence for **non-infill** football turfs is essentially absent. There are no robust datasets that describe how athletes load these surfaces in real tasks, how studs penetrate and micro-slip on dense carpets with thatch layers, or how these behaviours change with traffic and age.

A second, systemic gap is regulatory. Non-infill fields continue to be judged against test methods and pass/fail bands that were developed around infilled systems. Those limits have value for procurement and quality control, but they were not calibrated on the carpet–pad architectures that now replace loose performance infill. This misalignment makes product development harder—manufacturers must “design to” proxies that may not reflect true athlete loading—and it risks setting safety and performance precedents that are not evidence-based for non-infill designs.

This thesis responds to those gaps by focusing on non-infill surfaces. It proposes player-centred testing, complemented by targeted mechanical measures, to characterise traction, stud penetration, slip events, and impact response on carpet–pad systems. The goal is to generate baseline data for non-infill fields, evaluate how well current certification tests capture player-relevant behaviour, and indicate where adapted thresholds or additional metrics may be needed to support safe, performant, and manufacturable non-infill turfs.

3.9 Theoretical Framework

Core premise:

This study treats non-infill football turf as a distinct surface class with little player-based evidence. Existing limits and methods were developed on infilled 3G systems, and in-vivo datasets for non-infill are scarce[1], [2], [11], [12]. The framework is player-centred: the relevant loading is the short contact, small in-stance rotation, and rapid force rise that occur during shuttle-run turns. Athlete–surface behaviour will be quantified on three carpets without shock pads—new non-infill, worn non-infill, and 3G SBR-infilled—using Vicon motion capture and a force plate so surfaces tested in this study will not include shock pads. Variables include vertical and horizontal ground-reaction forces,Torque, utilised coefficient of traction, stance time and angle, and rearfoot–forefoot behaviour. Interpretation is anchored to task performance and slip control rather than peak values alone. Any method or threshold recommendations will be evidence-led: if non-infill supports the task at lower peaks and gentler stiffness, reporting a stiffness measure alongside peak torque and tighter rate control may be warranted; if not, the priority remains better logging without changing limits. The objective is to establish initial player-based ranges for non-infill and determine whether infill-derived interpretations require adjustment for this surface class.

4 METHODOLOGY

4.1 Research design

Three surface states were tested within a single session: a new non-infill carpet (GreenFields Pure PT), a worn non-infill carpet artificially prepared (3000 cycles artificially aged), and a third-generation (3G) infilled system (SIS Turf Xtreme Ultra) filled with silica sand and styrene–butadiene rubber (SBR) in that order. Only studded boots were used so that the measured behaviour reflected stud–carpet interactions rather than outsole-type contrasts.

4.1.1 Surface state standardisation:

For the 3G infilled surface, the field of play over the plate was raked and then rolled with a 20 kg roller fitted with stud protrusions to level the infill and set the playing face. For the non-infill carpets, the surface was raked before trials. As per the study protocol, the infill level on the 3G surface was maintained just below the fibre crowns and managed using the routine already described for player testing.

4.1.2 Trial structure, randomisation and acceptance rules

Each participant contributed ten valid right-foot contacts on each surface, yielding thirty analysed contacts per subject. Ten contacts per condition were chosen to provide a stable within-subject estimate while keeping exposure and fatigue within safe limits. Surface order was randomised independently for every participant to distribute learning and fatigue effects across conditions. A trial was accepted when a single right-foot contact occurred wholly within the force-plate boundaries, all required boot markers remained visible throughout stance, and the 180° change of direction was completed without a double-step on the plate. Attempts that failed these criteria were immediately re-attempted until the contact quota for that surface was achieved.

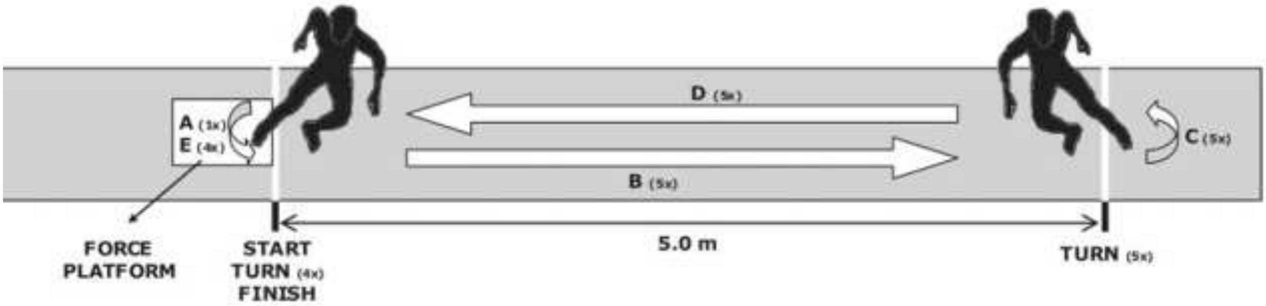


Figure 9 Shuttle runs Execution by participants[10]

4.1.3 Surfaces and preparation

This subsection defines the test surfaces and their preparation so the setup can be replicated. Gauge refers to tuft spacing. Dtex is fibre linear density in grams per 10,000 metres. Three surface states were mounted over the force-plate aperture. The approach and exit lanes stayed as the same non-infill carpet so the run-up was identical in all conditions.

The force-plate area measured **600 × 900 mm**. For each surface state, a pre-cut insert matching this planform area was seated **flush** to the surrounding lane. **White floor markers** were placed to cue the natural pivot point for the 180° turn. To prevent any carpet, slip under stud loading, the inserts were restrained with **industrial-grade double-sided tape applied to the backing**; no weights were used.

Non-infill (GreenFields Pure PT): A non-filled, three-eighths-inch gauge tufted synthetic grass with a 100 percent UV-resistant polyethylene monofilament pile made from two fibre types: ribbed rectangular, semi-texturized, about 160 micrometres and 6000/6 dtex; and diamond, LSR-texturised, about 150 micrometres and 8000/10 dtex. The combined designation is 14000/16 dtex. The carpet is tufted into a double polypropylene warp-knitted primary backing with a fibre-locked fleece of about 320 grams per square metre and finished with a latex secondary backing with drainage perforations of about 1,000 grams per square metre. Pile height 38 millimetres with a tolerance of plus or minus five percent; total thickness 40 millimetres with the same tolerance; stitch rate 105 per linear metre in width by 300 per linear metre in length and about 31,500 tufts per square metre; and about 378,000 main-pile filaments plus about 630,000 thatch filaments per square metre. Tuft bind is at least 40 newtons and water permeability is at least 500 millimetres per hour. Total product mass is about 5,164 grams per square metre. For testing, the carpet was restrained around its perimeter with double sided tape to prevent whole-sheet movement under stud loading.[13]

Worn non-infill. The same Pure PT product was used after a 3,000-cycle laboratory ageing protocol intended to emulate pile lay-over and partial backing exposure typical of sustained use.

3G infilled (SIS Turf Xtreme Ultra). A long-pile polyethylene monofilament system using a six-ply, 12,000 dtex propeller-shaped yarn on two woven polypropylene primary backings with a polyurethane coating of about 600 grams per square metre. Key specifications are Pile height 60 millimetres; pile weight 1,618 grams per square metre; machine gauge five-eighths of an inch; 17 stitches per 10 centimetres; and total product mass 2,434 grams per square metre. The playing system was filled with silica sand of 0.2 to 0.7 millimetres as the stabilising infill and SBR of 0.8 to 2.0 millimetres as the performance infill at recommended application rates of 25 kilograms per square metre for sand and 15 kilograms per square metre for SBR. During data collection, It was found after during the trials that the carpet was overfilled but since there was no shock pad the overfilled turf gave satisfactory values for AAA and RTT testing. the SBR level was maintained just below the fibre crowns and topped up after every two participants to keep vertical and rotational response consistent.

Only the turf seated over the force-plate aperture was exchanged between conditions; the carpet lane was unchanged. All overlays were brought flush to the surrounding lane and restrained at the perimeter of force plate to avoid steps at plate edges and to eliminate whole-carpet slip. For safety, participants ran only on carpeted lanes, and the concrete subfloor was never exposed under studs[41].

1 2 3

Figure 10. Test surfaces from left to right: (1) third generation (3G) turf, (2) worn non-infill turf, and (3) new non-infill turf.

Figure 11 surface prep for 3G turf

4.1.4 Footwear

All testing used a single studded model to remove footwear-design effects. Participants wore Puma Future 7 Ultimate FG/AG football boots in a ten-stud configuration suitable for firm natural grass and artificial turf. Pairs in UK sizes 6, 8 and 9 were available; each participant used the best-fitting size and then wore the same pair for all surface conditions. The stock NanoGrip sockliner was retained and no components were modified. Laces were secured to a consistent, firm tightness verified by the investigator before each block, and socks were standardised across sessions. This standardisation ensured that any between-condition differences reflected the surface states rather than changes in boot setup [42], [43].

Figure 12 Football studs used for the Study

4.1.5 Protocol control

Participants began shuttle-run blocks immediately after warm-up and marker checks. Participants received an audio countdown during each set to indicate the number of remaining repetitions. For safety, players were instructed to always remain within the carpeted lane, as slipping on exposed concrete with studs presents a high risk. Re-trials followed the acceptance rules already stated (off-plate, double-step, or marker dropout).

4.1.6 Participants

Five male football/rugby players(height, weight) volunteered and provided written informed consent under institutional ethical approval. All were free of lower-limb injury at the time of testing and familiar with artificial-turf play. Preparation consisted of five minutes of steady cycling on a Wattbike followed by static and dynamic lower limb stretches in the participants' own footwear. Test boots were then fitted, and comfort and sizing were verified prior to marker application.

Participant ID	Age (years)	Height (cm)	Weight (kg)	Left Leg Length (cm)	Right Leg Length (cm)	Left Knee Width (cm)	Right Knee Width (cm)	Left Ankle Width (cm)	Right Ankle Width (cm)
1	27	177.0	78.2	91.0	89.0	10.0	9.0	7.0	7.0
2	22	176.8	58.7	89.0	88.0	9.0	9.5	6.5	6.0
3	27	177.2	81.8	100.0	99.0	11.5	11.5	7.0	7.5
4	29	182.6	75.7	101.5	99.5	10.0	10.3	6.8	7.2
5	25	180.0	85.8	98.0	98.0	10.3	11.0	7.0	7.5

Table 4. Subject anthropometrics (height, mass, leg length, knee width)

4.1.7 Environment

All testing took place indoors to eliminate weather variation and to preserve a constant approach distance, camera coverage, and secure fixing of turf sections. The final protocol was shaped by inherent laboratory constraints—most notably the maximum available run-up of five metres, the footprint of the force plate, and the calibrated motion-capture volume—while still reproducing a football-relevant, worst-case traction demand.

4.2 Data collection methods

4.2.1 Motion Capture Setup and Calibration

Three-dimensional kinematics were recorded with a Vicon Vantage system sampling at 250 Hz. The cameras were setup in a way so they can successfully capture the turning point over the force plate covering that area into capture volume .six cameras were positioned on the medial side of the volume to enhance visibility for medial outsole markers and posterior Plug-in-Gait markers, with three cameras on the lateral side and three broadly downrange in the direction of travel. Pilot captures showed that this configuration reduced occlusions at the instant of pivot; several cameras were subsequently re-oriented to further reduce short gaps and the system was recalibrated after those adjustments.

Calibration followed the manufacturer workflow. A dynamic wand calibration established the common optical volume prior to data collection, after which static subject trials were captured to initialise auto-labelling and verify that all required markers were visible in the first and last frame of each capture. Calibration quality was monitored via Camera Calibration Feedback; the Vantage cameras exhibited image error in the range 0.15–0.22 pixels and world error typically below 0.35 mm, satisfying the laboratory acceptance criteria (image error < 0.25 px; world error < 0.40 mm). The global origin was set at a force-plate corner and a floor-plane procedure was performed; an observed 9 mm vertical offset between the reconstructed floor and the plate surface was corrected within Nexus so that the reconstructed plane coincided with the top of the plate. With this co-registration in place, kinematics and kinetics shared a common spatial frame in the exported files.

Force and motion were acquired concurrently in Nexus so that the 1000 Hz analogue channels from the force plate were written into the same C3D files as the 250 Hz trajectories. Cameras were positioned outside the running lane and pivot area; none were disturbed during testing.

Gap handling and quality control

After auto-labelling, marker data quality was reviewed in Vicon's Quality tab. Any brief occlusions within stance were reconstructed using the Rigid Body fill method: three intact markers from the same cluster were selected to re-estimate the missing marker based on rigid-body geometry. Trials with unreconstructable gaps within stance (i.e., where a continuous centroid trajectory could not be guaranteed) were rejected and immediately re-attempted. The target for accepted trials was no gaps within stance.

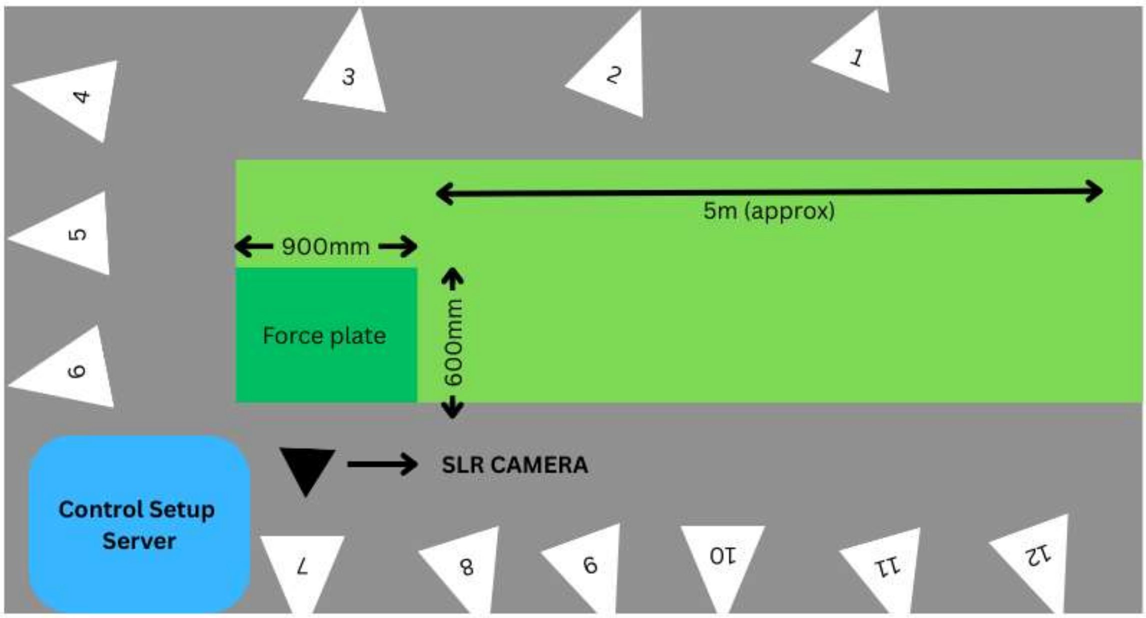


Figure 13.Lab layout: Vicon camera positions, lane setup, force plate, turf strips

4.2.2 Force Plate Configuration and Signal Conventions

Ground-reaction forces and moments were recorded with a six-component Kistler Type 9287 force plate at 1000 Hz and acquired within Vicon for export alongside the optical data. Throughout this thesis the laboratory global frame defined in Section 3.2.1 is used consistently: +Z is vertical (upwards), X is the approach direction, and Y is mediolateral. Forces are reported as (Fx, Fy, Fz) in newtons. The free moment Mz is the torsional moment about +Z at the centre of pressure and follows the right-hand rule (counterclockwise positive when viewed from above). Centre of pressure and Mz were computed from the plate channels using the world–plate alignment established during calibration, so kinetic variables are directly comparable with the kinematic trajectories without further spatial transformation after export. The plate was checked unloaded before data-collection blocks to verify baseline stability; the contact-detection threshold and other event definitions are specified in the Data processing section.

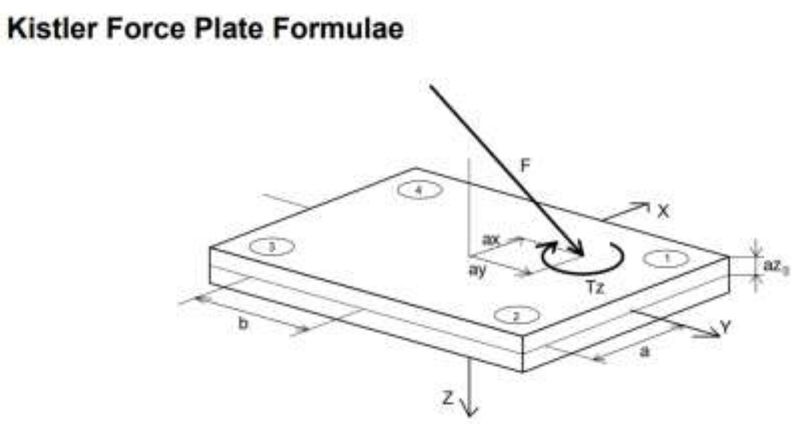


Figure 14.Force plate axes and free moment[44], [45].

4.2.3 Marker placement and segmentation

Boot kinematics at the shoe–surface interface were resolved by treating the boot as two rigid segments—forefoot and rearfoot—and tracking them with retro-reflective marker clusters bonded directly to the outsole. The forefoot segment carried six markers spaced across the distal stud rows and along the lateral–medial margin; the rearfoot segment carried five markers distributed across the heel stud array and posterior outsole. This layout provided geometric spread in three dimensions so that each cluster remained reconstruct able through the large rotations occurring during the 180° pivot, and it increased redundancy beyond the minimum three markers per segment to mitigate brief occlusions. The forefoot–rearfoot boundary was defined a priori by the separation between the two clusters on the outsole, and all segment calculations referenced these fixed cluster definitions rather than time-varying centre-of-pressure location.

Lower-body Plug-in-Gait anatomical markers were also applied to standard locations to support reliable auto-labelling and quality control within Nexus; however, the analysis reported in this thesis used only the right-boot clusters. Skin preparation included a light application of adhesive spray before placing anatomical markers to reduce dropout. Prior to dynamic trials, a static capture with all cluster markers visible in the first and last frames was recorded to initialise labelling templates. Marker integrity was checked between blocks, and any displaced markers were re-affixed before data collection resumed[46].

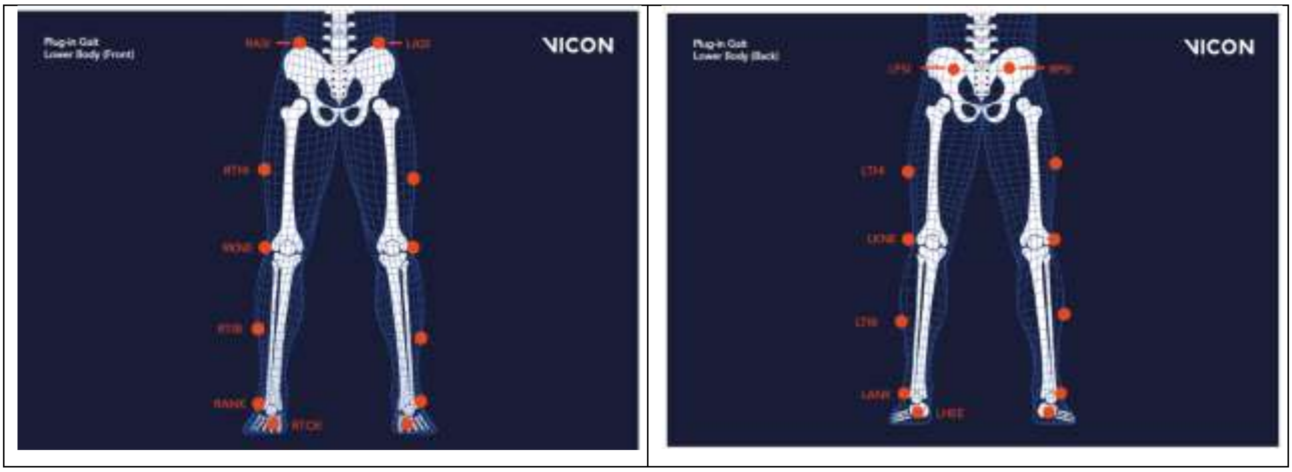


Figure 15.Lower body plugin gait markers locations

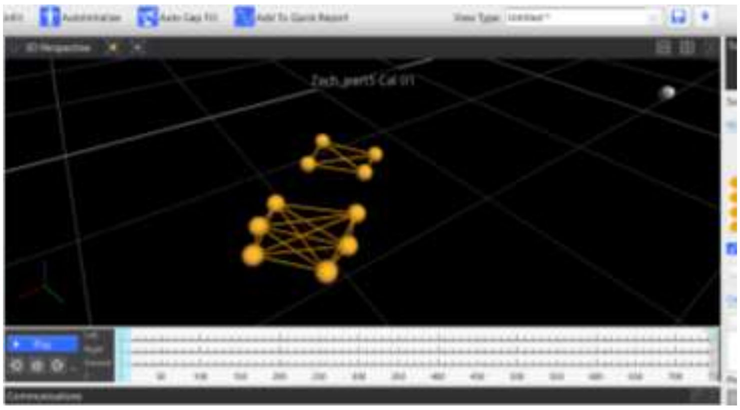


Figure 16 marker cluster used primarily for the study

4.2.4 Signal Processing and Event Detection

Processing steps were applied consistently across surfaces and participants. Trials were cropped to the stance phase of the right foot, with short marker gaps reconstructed and longer ones repeated. Force and kinematic data were low-pass filtered (zero-phase Butterworth) with pre-set cut-offs; event detection used unfiltered Fz to avoid bias. Initial contact was defined at $Fz \geq 20$ N.

Traction was calculated as the ratio of horizontal to vertical force ($\sqrt{(Fx^2 + Fy^2)}/Fz$), reported for the whole foot and by segment. Peak vertical force, horizontal force, and ground-reaction torque (Mz) were extracted within stance. Force–deformation behaviour was estimated from vertical force and displacement, with energy return taken from the hysteresis loop. All calculations used synchronised C3D files in a common spatial frame.

4.2.5 Force–Displacement Behaviour

The force–displacement curve can be derived from the vertical ground reaction force (GRFz). However, accurately determining the corresponding deformation remains challenging. Two potential approaches may be used:

1. Placing markers directly on the surface and estimating deformation from the vertical displacement of these markers.
2. Using the foot's initial contact and calculating deformation from the vertical movement of the segment centroid.

Both methods yield results, but with noticeable variation between them.

4.2.6 Traction Metrics

Definition and frame:

Coefficient of traction (CoT) was computed in the global laboratory frame (Z up, X approach, Y mediolateral) as the ratio of the horizontal to the vertical ground-reaction force. The horizontal resultant was formed from the force-plate channels as

$$Fh(t) \;=\; \surd(Fx(t)\text{^}2 \;+\; Fy(t)\text{^}2\,)$$

and the coefficient of traction over time was

$$CoT(t) \;=\; Fh(t)\;/\;Fz(t)$$

Stance was defined after De Clercq: initial contact was the first frame where Fz(t) ≥ 20 N, and toe-off was the last frame before Fz(t) < 20 N. Because CoT(t) varies most at the start and end of stance, the primary outcome was the average traction value, taken as the mean of CoT(t) from ten percent of stance until the instant when the vertical force first fell below one-half of body weight:

$$CoT_{plateau} = \;mean\;of\,(\;CoT(t)\;)\;for\;t\;\in\;[10\%\;of\;stance, first\;Fz(t)\;<\;0.5\;*\;BW]$$

Force channels (Fx, Fy, Fz) were low-pass filtered at 50 Hz with a zero-phase, fourth-order **Butterworth filter** and then decimated to 250 Hz to pair with kinematics. CoT(t) was calculated from these filtered forces. CoT is dimensionless, and one CoT_plateau value was obtained for every accepted contact; participant-level aggregation and statistical analysis are described later.

A separate rotational traction test (RTA) was performed before player trials to characterise the available traction of each surface state.

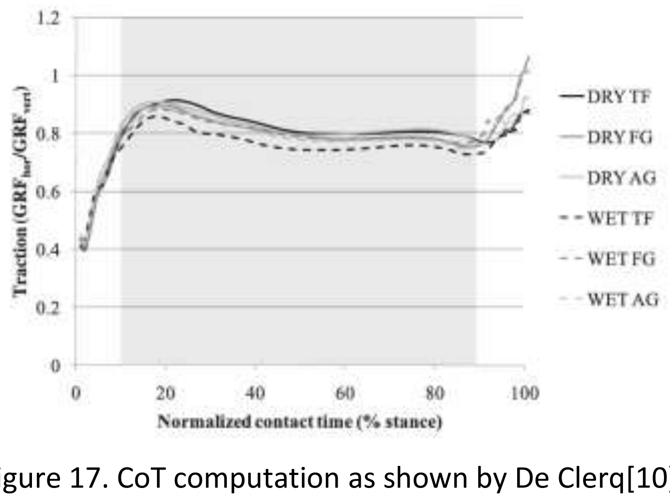


Figure 17. CoT computation as shown by De Clerq[10]

4.2.7 Mechanically Available Traction Testing (RTA Method)

Mechanically available traction was measured with the **Rotational Traction Athlete (RTA)** method so that player-executed traction could be interpreted against an independent ceiling. In dynamic tasks, utilised traction is typically lower than the mechanical limit measured by traction devices [5], [47], [48].

Apparatus and load. The RTA used a 150 mm circular test foot fitted with six standard football studs on a 46 mm pitch radius, mounted to a vertical shaft with low-friction support and a spring of 4 ± 1 N per millimetre. When the operator stood on the studded baseplate, the spring applied a **normal load of 450 ± 20 N** through the test foot. Integrated torque and angle sensors provided the measurements reported below [5].

Procedure:

Studs and disc were cleared of debris before each trial. The operator performed the prescribed balancing action to seat the baseplate and then applied a smooth rotation of at least 90 degrees in about four seconds, avoiding vertical load on the handle. Trials were accepted only when the mean rotational velocity from the start of rotation to peak torque lay between 20 and 50 degrees per second; trials outside this range were re-run [5]. Surfaces were brushed to a level condition before testing; measurements were taken indoors at standard room temperature.

Output variables

$$\text{T\_peak} = \text{maximum recorded torque during the rotation} \quad [\text{N} \cdot \text{m}]$$

$$\text{T\_10deg} = \text{torque recorded at } 10 \text{ degrees during the build} - \text{up} \quad [\text{N} \cdot \text{m}]$$

$$\text{RV\_mean} = (\theta\text{\_peak} - \theta\text{\_start}) \,/\, (\text{t\_peak} - \text{t\_start}) \quad [\text{degrees per second}]$$

$$\text{RSS} = \text{slope of torque versus angle between } 50\% \text{ and } 80\% \text{ of T\_peak} \quad [\text{N} \cdot \text{m per degree}]$$

Sampling and repeats:

Each surface state (new non-infill, worn non-infill, 3G infilled) was tested **three times** at distinct locations spaced at least 100 mm apart and from edges [5], [16]. For each surface, the **mean of the three device readings** was reported. Runs were discarded only if the rotational-velocity criterion was not met or if the device trace showed an obvious artefact; in such cases the trial was repeated.

Surface Type	Max Torque (Nm)	Shear Stiffness (Nm/deg)	T10 (Nm)	RVmn
SBR Infill Turf	45.6	1.3	20.6	38.3
PurePT Fresh	26.7	0.45	15.8	45
PurePT 3000 Wear	36.4	0.62	16.5	39.7

Table 5 Rotational traction apparatus (RTA) results for one participant 5 across three surface conditions. Data shown are mean values of three trials per surface

4.3 Sampling techniques

4.3.1 Sampling strategy

Participants were deliberately selected to match the demands of the high-traction task and ensure biomechanical consistency under controlled conditions. Rather than recruiting randomly, the study targeted individuals with a sporting background in football or rugby, capable of executing sharp directional changes while wearing studded footwear. This approach ensured that all participants had the physical ability and movement familiarity required to safely and reliably perform the 180° pivot task on all surface types.

4.3.2 Inclusion criteria

The sample included five right-foot dominant, injury-free male athletes aged 18–30, all actively engaged in football or rugby at a university or recreational level. Prior experience on artificial turf was required to minimise surface unfamiliarity and promote realistic loading behaviour. Participants were fitted with standardised footwear from a controlled size range (UK 6, 8, 9). Full-body morphology was recorded, including lower-limb segment dimensions, to support marker placement and provide context for interpreting force and displacement patterns. Heights ranged from 176.8 cm to 182.6 cm, and body masses from 58.7 to 85.8 kg, offering a representative sample of recreational regular players while preserving control over technique consistency.

4.3.3 Sample size justification

Although the study initially aimed to recruit seven plus participants, laboratory constraints plus timings and availability limited the final sample to five. However, each individual contributed thirty valid trials (ten per surface), generating a high-density dataset suitable for within-subject comparisons. This design is consistent with prior biomechanics studies that prioritise controlled, repeated measures over population generalisability. The dataset supports detection of directional trends across surface types even with low absolute sample size, particularly when using non-parametric statistics designed for small-n, repeated-trial designs.

4.3.4 Generalisability and limitations

The restricted sample—limited to male, right-foot dominant university athletes—limits direct extrapolation to female players, left-dominant individuals.
Furthermore, the lab-controlled environment excluded field conditions such as surface moisture or compaction variability. However, this homogeneity was intentional: it allowed the study to isolate the effect of surface condition under tightly constrained and repeatable loading scenarios. The resulting dataset offers a foundational reference for future biomechanical evaluations of non-infill turf under more ecologically valid conditions.

4.4 Data analysis and procedures

Kinetic (1000 Hz) and kinematic (250 Hz) data were recorded synchronously in Vicon Nexus and exported in a common laboratory frame (+Z vertical, +X approach, +Y mediolateral). Forefoot and rearfoot outsole clusters were labelled; brief within-stance occlusions were reconstructed using a rigid-body fill.marker trajectories were filtered at a lower cut-off to preserve event timing. Forces were then resampled to 250 Hz so kinetics and kinematics shared a common time base. **Initial contact (IC)** was the first sample where vertical force (Fz) reached ≥ 20 N; **toe-off (TO)** was the last sample before Fz fell < 20 N. All time-series outcomes were computed within this stance window and re-expressed on a **normalised 1–100 % stance** scale to align force, torque, displacement and angle frame-by-frame.

Free moment:

to quantify rotational demand at the shoe–surface interface, the vertical ground-reaction torque (free moment) about +Z at the centre of pressure (CoP) was obtained from the force plate as

$$Tz = Mz - Fy \times ax + Fx \times ay \quad (N \cdot m) \quad (1)$$

Free moment or torque was attempted to be quantified with varying level of success

Peak vertical force (N)

Defined as the maximum Fz within stance on the filtered trace. For comparability across participants, values were also expressed per body mass ($N \cdot kg^{-1}$) where appropriate.

Peak horizontal force (N)

Computed from the laboratory-plane resultant of the horizontal force components; the peak is the maximum value within stance. (Where helpful, braking and propulsive peaks may be described by the sign of the fore–aft component, but the primary outcome is the resultant peak.)

Rate of force development (RFD

RFD was reported as the slope of the peak reaction forces in the weight acceptance segment(0-20%)

Coefficient of traction (CoT)

CoT_plateau was computed using the method described in Traction Metrics previously, based on De Clercq definition.

Aggregation for statistics

Each participant could provide ten valid contacts per surface. All variables above can be computed per contact on the stance-normalised data and then reduced to a single representative value per participant per surface (mean or median, pre-specified per variable) for within-subject comparisons. Body-mass-normalised forms ($N \cdot kg^{-1}$) were reported where appropriate.

4.4.1 Statistical analysis

All statistical analyses were conducted in Python. The study followed a **within-subject repeated-measures design**, as each participant completed the pivoting task under all three surface conditions. Additional **between-subject comparisons** were performed to assess variability across participants and support surface-specific conclusions. Given the small sample size (n = 5), **non-parametric tests** were prioritised. Parametric sensitivity analyses were conducted where appropriate.

For each surface condition, 10 trials per player were recorded. For all statistical comparisons, data were first **collapsed to a single representative value per player per condition**, using either the **median** or **mean** of the 10 trials depending on within-condition variability.

The **Friedman test(non-parametric equivalent for ANNOVA)** was used as the **primary test** for each dependent variable across the three surfaces.

When the Friedman test was significant (α = 0.05), **Wilcoxon signed-rank tests** were conducted for **pairwise comparisons** (New NI vs Worn NI, New NI vs 3G, Worn NI vs 3G), with **Holm–Bonferroni correction** applied for multiple testing[10].
To assess whether the same surface effects were observed across players:
The **Kruskal–Wallis test** was used to compare player-level medians on a given surface. This exploratory step helped assess whether players loaded the same surface differently.
Lack of significant between-player variation will support the interpretation that surface effects, rather than inter-individual differences, were the primary source of variation.
Where appropriate, **body mass–normalised values** (e.g., N/kg for force, Nm/kg for torque) were used to minimise anthropometric bias across subjects.
To assess the consistency of repeated trials within a surface condition:
Intraclass correlation coefficients (ICC) and **coefficient of variation (CV%)** were calculated for each dependent variable, across the 10 repetitions within each condition.
Values of **ICC ≥ 0.75** and **CV ≤ 15%** were interpreted as evidence of acceptable within-session repeatability [49].This ensured that collapsing trials into a single representative value per condition was justified.
Interpretation and Threshold Adjustments
Observed surface-specific differences in key outcome variables (e.g., torque, traction ratio, shock absorption) were interpreted with reference to the **threshold criteria** defined in current mechanical testing standards such as the FIFA AAA and RTA protocols. If in-situ values on non-infill surfaces consistently **exceeded or fell below** the standard thresholds, this was used as evidence to suggest **modifications to existing test specifications** or **threshold limits**.

4.5 Ethics and safety

The study received a favourable ethical opinion from the institutional Ethics Review Sub-Committee for the project titled **Boot-Surface Interaction** (Project ID 21391; reference 2025-21391-22196). The letter confirms that the protocol has a favourable opinion through 31 December 2026, that amendments should be submitted if the design changes substantially, and that transitions to online data collection do not require an amendment unless such changes materially alter the study. Any incident with adverse effect must be reported to the Sub-Committee using the Adverse Events Report within the LEON system, and the protocol follows institutional guidance on the collection of sex and gender data. All participants provided informed consent after a full explanation of the study aims, procedures and risks, and they were free to withdraw at any time without consequence.

All running occurred on turf overlays that were firmly fixed to the laboratory floor; the concrete substrate was never used for locomotion under studs because of elevated slip risk. The testing team monitored execution in real time and terminated a trial immediately if a safety concern arose. Rest intervals were enforced between blocks to minimise fatigue. Footwear fit was confirmed prior to testing, and the booted foot was the only contact on the force plate during valid trials, avoiding double-step events that might compromise stability or data quality.

5 Results and analysis

5.1 Presentation and findings

5.1.1 RTA for every participant

Participant	Surface Type	Max Torque (Nm)	Shear Stiffness (Nm/deg)	T10 (Nm)	RVmn
1	SBR Infill Turf	45.57	1.3	20.6	38.33
	PurePT Fresh	26.67	0.45	15.83	45
	PurePT 3000 Wear	36.43	0.62	16.53	39.67
2	SBR Infill Turf	44.9	1.27	19.47	44.67
	PurePT Fresh	24.97	0.56	17.2	52
	PurePT 3000 Wear	43.23	0.77	16.57	41
3	SBR Infill Turf	40.87	1.06	18.5	49.33
	PurePT Fresh	25.57	0.5	16.1	58
	PurePT 3000 Wear	37.4	0.6	17.27	55
4	SBR Infill Turf	40.59	1.08	18.33	53.33
	PurePT Fresh	25.8	0.48	16.2	62
	PurePT 3000 Wear	36.3	0.52	15.7	57.33
5	SBR Infill Turf	41	1.08	19.43	43.67
	PurePT Fresh	25.5	0.65	18.1	49
	PurePT 3000 Wear	39.4	0.69	16.97	45

Figure 18 RTA data

The RTA results show clear differences between surfaces. SBR infill turf generated the highest maximum torque and shear stiffness, meaning it offered the greatest resistance to rotational movement. Fresh PurePT consistently produced the lowest torque values, with less stiffness but higher rotational velocities, reflecting a more compliant surface. Worn PurePT (3000 wear) displayed greater torque than the fresh sample, moving closer to SBR levels, which suggests that surface ageing increased resistance to rotation.

5.1.2 AAA findings

Participant	Surface	VD (mm)	FR (%)	ER (%)	v impact (m/s)	v rebound (m/s)
1	PT Fresh	8.51	31.3	40.9	1.17	-0.74
	PT3000	9.13	37.6	37.7	1.13	-0.69
	SBR	9.77	59.2	41.9	1.08	-0.7
2	PT Fresh	8.63	30.6	40.9	1.18	-0.75
	PT3000	9.33	34.2	38.6	1.17	-0.73
	SBR	9.25	59.4	42.2	1.04	-0.68
3	PT Fresh	20.12	37.7	56.4	1.51	-1.15
	PT3000	13.11	34.4	7.8	1.45	-0.4
	SBR	22.08	58.8	18.3	1.41	-0.59
4	PT Fresh	8.72	31.6	40	1.18	-0.75
	PT3000	9.01	33.9	39.2	1.16	-0.73
	SBR	8.85	56.1	43.9	1.05	-0.7

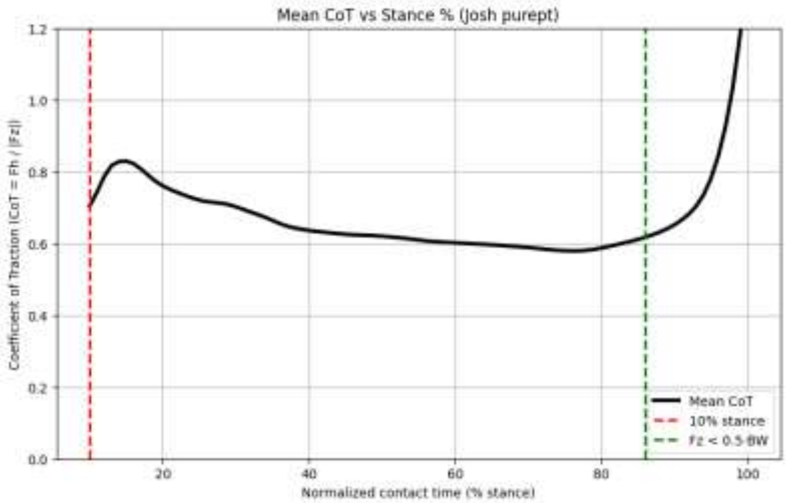
Table 6 AAA data

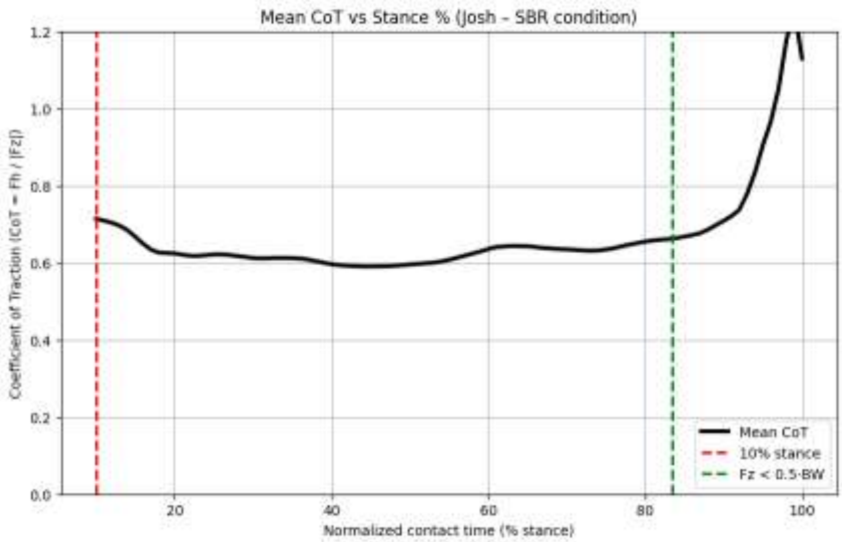
The AAA results highlight strong surface effects. SBR infill consistently showed the highest force reduction (≈56–59%) across participants, alongside moderate energy return (≈42%) and slightly lower rebound velocities, indicating greater absorption and damping. Fresh PurePT surfaces had the lowest force reduction (≈30–32%) but the highest energy return (≈40–56%), making them more elastic and spring-like. Worn PT3000 surfaces generally showed intermediate values, with force reduction higher than fresh PT but lower than SBR, and energy return values reduced compared to fresh. Notably, Harry's data showed larger variability, with very high deformation on SBR and fresh PT, suggesting participant-specific interaction effects.

5.2 Player movement study results

1 Coefficient of traction (CoT)

PurePT Fresh provided greater effective traction than SBR across the cut. The CoT plateau—our primary "usable traction" metric—was 0.836 on Fresh vs 0.792 on SBR, and peak CoT in weight acceptance (0–20% stance) was 1.000 ± 0.061 vs 0.932 ± 0.053. In practical terms, athletes could generate and sustain a slightly higher horizontal force for a given vertical load on the non-infill surface, which supports tighter redirection with less reliance on sliding. Because this advantage appears in the early stance phase, it is directly relevant to braking and pivot initiation.





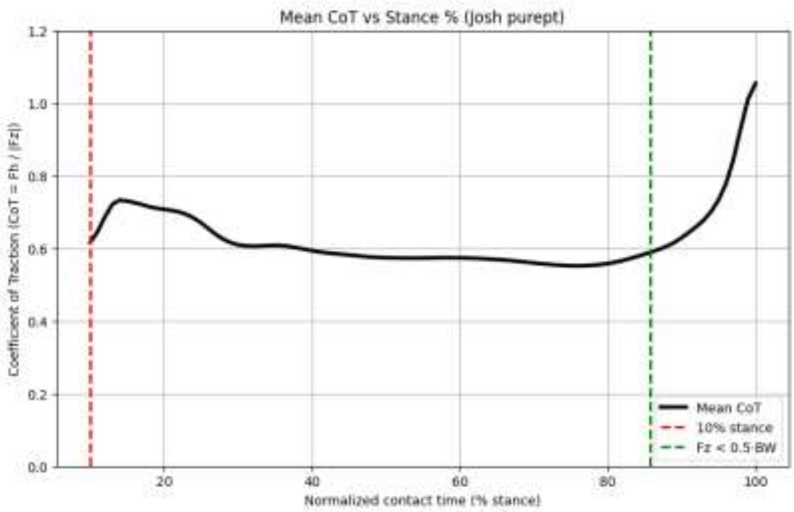


Figure 19 PARTICPANT 3 results CoT for different surfaces

CoT_plateau calculated from the 10% stance to when vertical force drops below half the bodyweight showed consistency across trials for different participants .

Participant	Surface	CoT_plateau
1	SBR	0.627
	PurePT Fresh	0.648
	PurePT Worn	0.605
2	SBR	0.838
	PurePT Fresh	(open for reevaluation)1
	PurePT Worn	0.84
3	SBR	0.792
	PurePT Fresh	0.836
	PurePT Worn	0.848

Table 7 Cot_plateue for three participants

2 Rate of force development (RFD)

Vertical loading developed slightly faster on Fresh non infill (52.99 ± 4.63 kN/s) than infill (50.73 ± 10.70 kN/s), and with less variability, suggesting a more consistent vertical response. Conversely, horizontal RFD was a touch higher on infill (34.43 ± 7.55 kN/s) than non infill(32.76 ± 2.93 kN/s), indicating shear forces ramped a bit more abruptly on the infill system. Together, these results imply new non infill turf delivers a quick, consistent vertical “set” for the foot, while SBR encourages a slightly sharper rise in shear—an effect that also aligns with its larger backward slip.

We also see a peak in Cot when the horizontal and the vertical forces are at their peak which also makes sense .

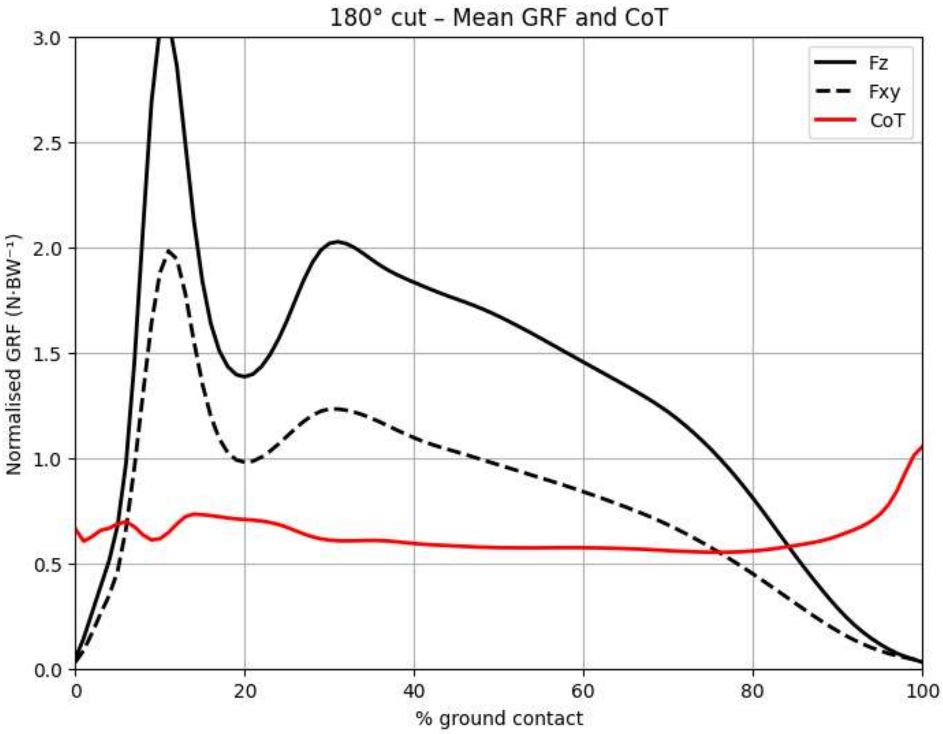


Figure 20 normalised GRF for non infill worn

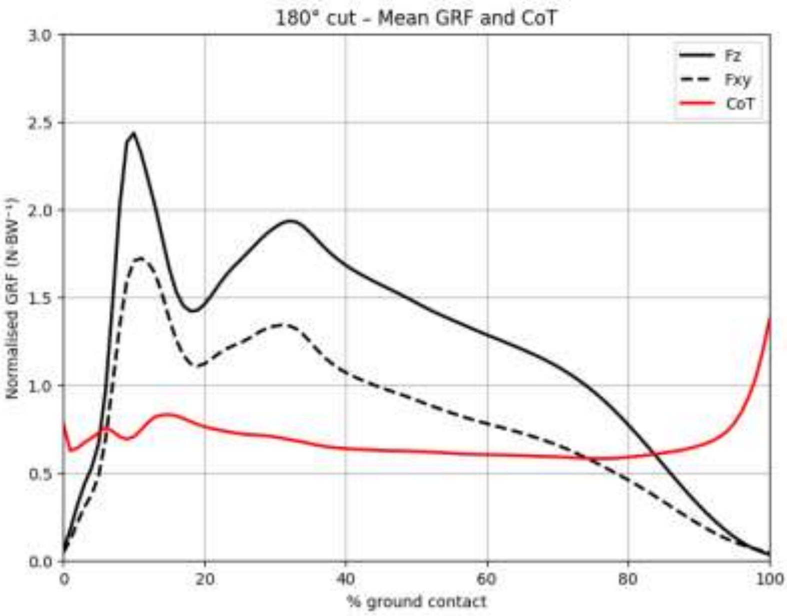


Figure 21 GRF 's body weight normalised noninfill new

3 Peak forces

Peak vertical forces were near-identical between surfaces—1800.7 N (≈2.14×BW) on SBR and 1789.9 N (≈2.13×BW) on Fresh—so the impact demand of the 180° cut was essentially unchanged by surface type. Peak horizontal forces were slightly higher on Fresh (1435.4 N; ≈1.71×BW) than SBR (1390.8 N; ≈1.65×BW), consistent with the higher CoT on Fresh. Importantly, Fresh allowed greater horizontal force production without elevating vertical peaks, suggesting improved grip did not come with extra impact loading.

Participant	Surface	Fz_peak (N)	Fz_peak / kg	Fz_peak / BW	Fh_peak (N)	Fh_peak / kg	Fh_peak / BW
1	SBR	1993.07	24.37	2.48	1427.73	17.45	1.78
	PurePT Fresh	2059.06	25.17	2.57	1434.86	17.54	1.79
	PurePT Worn	2728.16	33.35	3.4	1725.51	21.09	2.15
2	SBR	1507.3	19.91	2.03	1392.5	18.4	1.88
	PurePT Fresh	1410.8	18.64	1.9	1389.7	18.36	1.87
	PurePT Worn	1522.5	20.11	2.05	1415.9	18.7	1.91
3	SBR	1800.71	22.01	2.14	1390.83	16.21	1.65
	PurePT Fresh	1789.95	20.86	2.13	1435.35	16.73	1.71
	PurePT Worn	1723.57	20.09	2.05	1478.79	17.24	1.76

Table 8 table showing peak forces of 3 different participant

4 Engagement of Forefoot vs Rearfoot

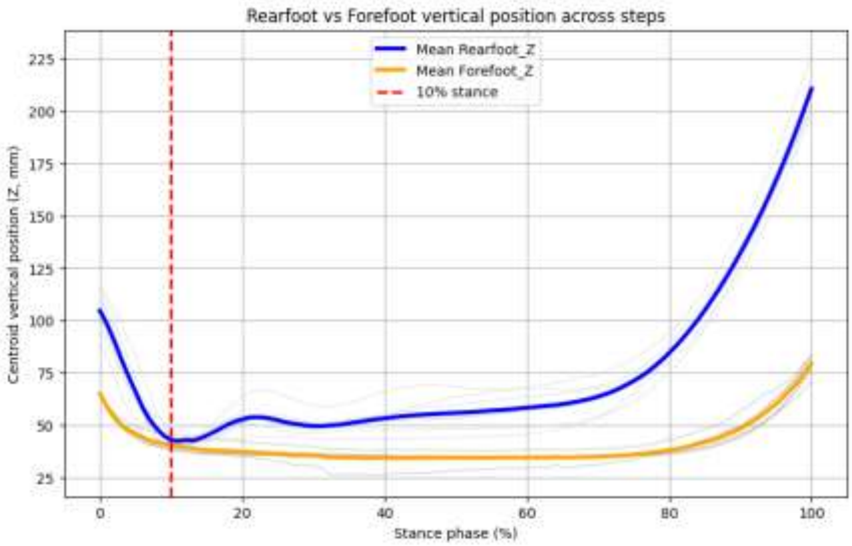


Figure 22 engagement of segments of feet with the ground

The forefoot engaged the surface in a shuttle run across all trials regardless of the surface .

5 Slippage (rearfoot A-P)

The forefoot centroid anterior–posterior displacement curve (Figure 23) showed a rapid increase in the first 10–15% of stance, with displacements rising to approximately 40–60 mm immediately after initial contact. This reflected the brief forward translation of the forefoot segment as the foot loaded and braking forces were applied. Following this initial rise, displacement values plateaued through mid-stance (≈45–55 mm), indicating that once the studs engaged with the turf, little further anterior movement occurred and the foot was held relatively stable. Toward the end of stance (80–95%), displacement declined gradually before dropping sharply at toe-off, leaving a net IC–TO forefoot translation of around 20–25 mm on average. While some steps showed larger excursions (up to >60 mm), the overall pattern was consistent: the majority of functional slip occurred immediately after contact, after which forefoot position remained largely fixed until unloading.

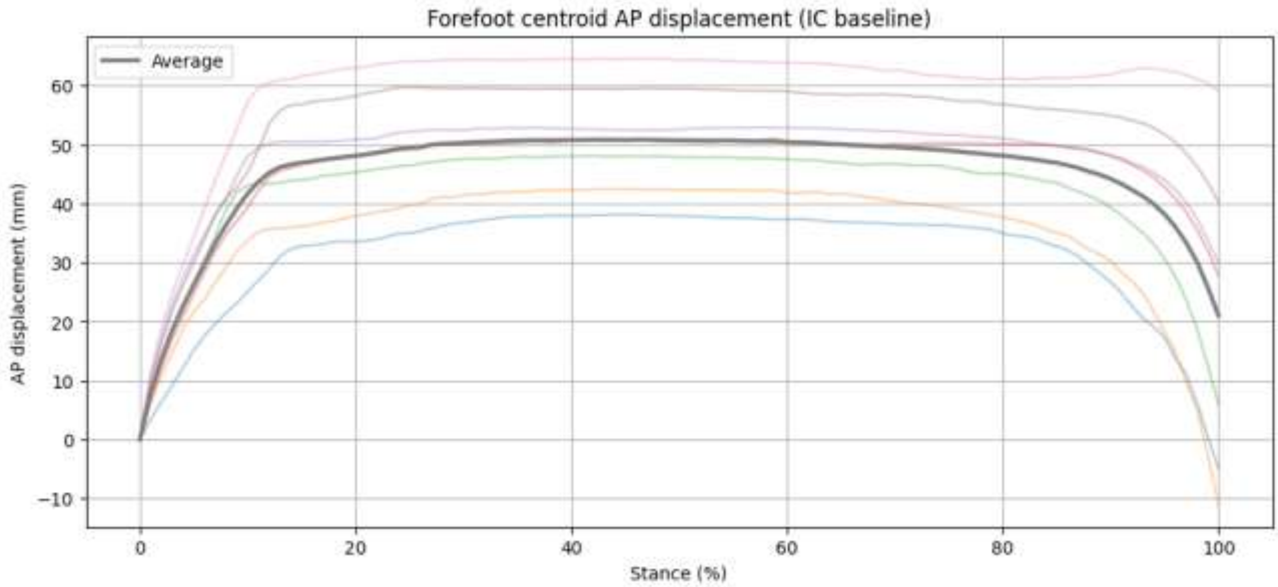


Figure 24 purept3000 worn anterior posterior displacement of front foot

6 Discussion

The in-vivo findings enable comparison of artificial turf systems by considering peak forces, rate of force development (RFD), and coefficient of traction (CoT). The SBR infill system provided stable but slightly lower traction, the Fresh PurePT non-infill surface offered greater grip while maintaining similar vertical loading, and the worn PurePT condition shifted responses towards higher force peaks and faster vertical loading rates.

Vertical peak forces were broadly comparable between SBR and Fresh PurePT, typically falling in the range of 2.0–2.5 × BW, indicating that vertical impact demand was not strongly surface-dependent. Differences were more evident in the horizontal forces, where Fresh PurePT consistently supported slightly higher peaks (1.7 × BW) compared to SBR (1.65 × BW). This demonstrates that the non-infill surface provided greater horizontal grip without amplifying vertical impact. By contrast, the worn PurePT condition produced the largest vertical and horizontal peaks, suggesting that ageing increased resistance to redirection movements and altered the overall loading profile.

Surface effects were clearer in loading rates. Fresh PurePT produced slightly higher and more consistent vertical RFD (53 kN/s) compared to SBR (51 kN/s), indicating a sharper but stable vertical response. Worn PurePT produced the steepest vertical loading rates (up to 63 kN/s), implying that surface ageing stiffens the system and accelerates impact onset. For horizontal RFD, SBR generally showed higher values (34 kN/s) than Fresh PurePT (33 kN/s), reflecting a sharper rise in shear loading on infill systems. Together, this suggests that Fresh non-infill turf supports a quick but stable vertical set, SBR encourages sharper shear loading, and worn surfaces intensify vertical loading.

Cot_plateau value was in between (0.6 – 0.85) for every surface giving a clear value for traction metrics.

6.1 Limitations

Recruitment for this study proved challenging, which restricted the participant pool and limited the scope for statistical testing. Several trials had to be discarded due to participants presenting with unorthodox running styles, double contacts on the force plate, or marker dropout, particularly during the first participant's session where early detachment reduced usable data.

Timeline for analysis was also pushed because of multiple injuries from earlier participants sustained pushing data capture timeline further.

These factors reduced the overall dataset and meant that statistical comparisons which was set out earlier between surfaces could not yield reliable significance levels. The timing of data collection also constrained outcomes, as the human trials were conducted later than planned; an earlier schedule would have allowed for more repeat trials and improved data balance across participants. Attempts were made to evaluate additional parameters such as torque, deformation, and rotational behaviour of the foot, but these analyses did not produce stable results and were therefore excluded, narrowing the study's ability to meet its full objectives. Furthermore, scrutiny of the algorithms used to calculate force- and traction-based variables should be undertaken more rigorously in future work to ensure robustness and reproducibility. Finally, while deformation was approximated indirectly, direct capture through enhanced motion-capture protocols could provide more realistic estimates of surface compliance, strengthening ecological validity and aligning results more closely with true player–surface interaction.

Out of the six objectives set for this study, the first three were completed successfully, including the literature review, identification of key biomechanical parameters, and the capture and analysis of in-vivo player data on non-filled turf. The remaining objectives—comparison with mechanical standards, cross-surface performance evaluation, and formulation of refined recommendations—were only partially achieved. The main bottleneck was the complexity of data analysis, where variability and noise in the collected signals led to inconclusive results that limited the scope of direct comparisons and broader recommendations.

6.2 Implications for mechanical testing

The results highlight limitations of current mechanical standards in representing real player loading.

- RTA: Protocols apply 460 N vertically, whereas player peaks exceeded 2.5 × BW (~2,700 N) vertically and 2.15 × BW (~1,730 N) horizontally. Improved tests should combine higher vertical loads with realistic shear forces.
- AAA: Current focus on vertical shock absorption and energy return does not reflect the combined vertical and horizontal forces observed. Testing should reproduce 2.0–2.5 × BW vertical peaks with 1.7–2.1 × BW horizontal shear and realistic rise times.

7 Conclusion

This project set out to investigate boot–surface interaction on non-infill artificial turf using in-vivo biomechanical testing and to compare the outcomes against the assumptions embedded within current mechanical test standards. Although challenges in participant recruitment and data loss limited the overall dataset, the study was able to capture meaningful differences between SBR infill, fresh non-infill, and worn non-infill surfaces in terms of peak forces, loading rates, and traction behaviour. Worn non-infill turf shifted responses further, with steeper vertical loading rates and higher combined force peaks, indicating that surface ageing alters playability in ways not captured by current mechanical standards.

These findings highlight some implications. Non-infill surfaces can provide comparable or even superior traction performance to infill systems, but their behaviour changes with wear, underlining the importance of long-term evaluation, Player loading .

Overall, this study provides initial biomechanical evidence that in-vivo testing for non infill turfs .

8 Acknowledgements

I would like to express my sincere gratitude to **Dr. Steph Forrester** for her invaluable guidance and support throughout this project. Her feedback at each stage was instrumental in shaping the study and ensuring its successful completion. I am also deeply thankful to **Utpal Narvekar**, PhD candidate, for his assistance during the study, particularly with the Vicon system and data collection. Finally, I would like to thank **PUMA** for kindly supplying the footwear used in the testing, without which this research would not have been possible.

9 REFERENCES

[1] D. C. Cole, "Numerical Modelling of 3G Artificial Turf Under Vertical Loading," 2020.

[2] P. Fleming, "Artificial turf systems for sport surfaces: Current knowledge and research needs," *Proc Inst Mech Eng P J Sport Eng Technol*, vol. 225, no. 2, pp. 43–64, 2011, doi: 10.1177/1754337111401688.

[3] S. Forrester and P. Fleming, "Traction forces generated during studded boot-surface interactions on third-generation artificial turf: A novel mechanistic perspective," *Engineering Reports*, vol. 1, no. 5, Dec. 2019, doi: 10.1002/eng2.12066.

[4] H. P. Gould, S. J. Lostetter, E. R. Samuelson, and G. P. Guyton, "Lower Extremity Injury Rates on Artificial Turf and Natural Grass Playing Surfaces: A Systematic Review," *Foot Ankle Orthop*, vol. 7, no. 1, Jan. 2022, doi: 10.1177/2473011421s00217.

[5] "TEST MANUAL I: TEST METHODS FIFA QUALITY PROGRAMME FOR FOOTBALL TURF."

[6] D. Loud, P. Grimshaw, R. Kelso, and W. S. P. Robertson, "Effect of Soccer Boot Outsole Configuration on Translational Traction Across Both Natural and Artificial Playing Surfaces," *Orthop J Sports Med*, vol. 12, no. 8, Aug. 2024, doi: 10.1177/23259671241259823.

[7] H. McGowan, P. Fleming, D. James, and S. Forrester, "Understanding the variability in rotational traction testing on artificial turf," *Sports Engineering*, vol. 26, no. 1, Dec. 2023, doi: 10.1007/s12283-023-00426-4.

[8] ARCHIE NEWMAN, "Wolfson School of Mechanical, Electrical and Manufacturing Engineering Individual Project Final Report Project title: Measuring studded footwear-surface interaction," 2024.

[9] "Wolfson School of Mechanical, Electrical and Manufacturing Engineering Individual Project Final Report Project title: Measuring studded footwear-surface interaction."

[10] D. De Clercq, G. Debuyck, J. Gerlo, S. Rambour, V. Segers, and I. Van Caekenberghe, "Cutting performance wearing different studded soccer shoes on dry and wet artificial turf," *Footwear Sci*, vol. 6, no. 2, pp. 81–87, 2014, doi: 10.1080/19424280.2014.895056.

[11] H. Driscoll, J. Kelley, B. Kirk, H. Koerger, and S. Haake, "Measurement of studded shoe-surface interaction metrics during in situ performance analysis."

[12] D. De Clercq *et al.*, "3 Boot-turf interaction during a 180° cutting movement on artificial turf when wet and dry."

[13] "Pure PT | GreenFields Europe." Accessed: Aug. 03, 2025. [Online]. Available: https://www.greenfields.eu/products/non-infill/Pure-PT

[14] "In-fill Artificial Grass vs non in-fill, whats the difference? - Trulawn Artificial Grass." Accessed: Aug. 03, 2025. [Online]. Available: https://www.trulawn.co.uk/customer-support/faq/in-fill-and-non-in-fill-grass/

[15] "Non infill vs infill artificial grass. Which should I choose? - Forever Green Lawns." Accessed: Aug. 03, 2025. [Online]. Available: https://www.forevergreenlawns.co.uk/faq/infill-vs-non-infill-artificial-grass/

[16] “TEST MANUAL II: TEST REQUIREMENTS FIFA QUALITY PROGRAMME FOR FOOTBALL TURF 2024 EDITION FIFA Quality Programme for Football Turf.”

[17] A. Thomson, R. Whiteley, M. Wilson, and C. Bleakley, “Six different football shoes, one playing surface and the weather; Assessing variation in shoe-surface traction over one season of elite football,” *PLoS One*, vol. 14, no. 4, Apr. 2019, doi: 10.1371/journal.pone.0216364.

[18] “Which Shoes Should You Use On Artificial Grounds? - On The Line.” Accessed: Aug. 08, 2025. [Online]. Available: https://www.lovellsoccer.co.uk/blog/post/which-shoes-should-you-use-on-artificial-grounds

[19] D. McGhie and G. Ettema, “Biomechanical analysis of traction at the shoe-surface interface on third-generation artificial turf,” *Sports Engineering*, vol. 16, no. 2, pp. 71–80, Jun. 2013, doi: 10.1007/s12283-013-0115-1.

[20] D. Cole *et al.*, “Comparison of player perceptions to mechanical measurements of third generation synthetic turf football surfaces,” *Sports Engineering*, vol. 26, no. 1, Dec. 2023, doi: 10.1007/s12283-022-00398-x.

[21] C. Webb, S. Forrester, and P. Fleming, “Rotational traction behaviour of artificial turf,” in *Procedia Engineering*, Elsevier Ltd, 2014, pp. 853–858. doi: 10.1016/j.proeng.2014.06.144.

[22] “Advanced Artificial Athlete (AAA Or Triple A) For Measuring Sports Surfaces.” Accessed: Aug. 19, 2025. [Online]. Available: https://labosport.com/advanced-artificial-athlete-aaa-or-triple-a/

[23] D. Cole, P. Fleming, K. Morrison, and S. Forrester, “Evaluation of the advanced artificial athlete and Hall effect sensors for measuring strain in multi-layer sports surfaces,” *SN Appl Sci*, vol. 2, no. 3, Mar. 2020, doi: 10.1007/s42452-020-2178-5.

[24] C. H. Webb, “MECHANICAL TRACTION BEHAVIOUR OF ARTIFICIAL TURF,” 2016.

[25] H. McGowan, P. Fleming, J. H. Pak, D. James, and S. Forrester, “The effect of rotational velocity on rotational traction across a range of artificial turf surface systems,” *Sci Rep*, vol. 13, no. 1, Dec. 2023, doi: 10.1038/s41598-023-48134-0.

[26] R. El Kati, “EFFECT OF MECHANICAL BEHAVIOUR OF ARTIFICIAL TURF ON PLAYER-SURFACE INTERACTION IN SOCCER,” 2012.

[27] M. Ferrandino, S. Forrester, and P. Fleming, “The player surface interaction of rugby players with 3G artificial turf during rugby specific movements,” in *Procedia Engineering*, Elsevier Ltd, 2015, pp. 308–313. doi: 10.1016/j.proeng.2015.07.249.

[28] R. F. Kirk, I. S. G. Noble, T. Mitchell, C. Rolf, S. J. Haake, and M. J. Carré, “High-speed observations of football-boot-surface interactions of players in their natural environment,” *Sports Engineering*, vol. 10, no. 3, pp. 129–144, Sep. 2007, doi: 10.1007/BF02844185.

[29] L. Damm, V. Stiles, and S. Dixon, “Human-shoe-surface interaction.” [Online]. Available: https://www.researchgate.net/publication/288668061

[30] R. Kaila, “Influence of modern studded and bladed soccer boots and sidestep cutting on knee loading during match play conditions,” *American Journal of Sports Medicine*, vol. 35, no. 9, pp. 1528–1536, Sep. 2007, doi: 10.1177/0363546507300257.

[31] B. M. Nigg, D. J. Stefanyshyn, A. I. Rozitis, and A. Mündermann, “Resultant knee joint moments for lateral movement tasks on sliding and non-sliding sport surfaces,” *J Sports Sci*, vol. 27, no. 5, pp. 427–435, 2009, doi: 10.1080/02640410802669161;JOURNAL:JOURNAL:RJSP20;REQUESTEDJOURNAL:JOURNAL:RJSP20;PAGE:STRING:ARTICLE/CHAPTER.

[32] V. H. Stiles, I. T. James, S. J. Dixon, and I. N. Guisasola, “Natural turf surfaces: The case for continued research,” *Sports Medicine*, vol. 39, no. 1, pp. 65–84, 2009, doi: 10.2165/00007256-200939010-00005,.

[33] E. M. Zanetti, C. Bignardi, G. Franceschini, and A. L. Audenino, “Amateur football pitches: Mechanical properties of the natural ground and of different artificial turf infills and their biomechanical implications,” *J Sports Sci*, vol. 31, no. 7, pp. 767–778, Apr. 2013, doi: 10.1080/02640414.2012.750005.

[34] M. R. Shorten, “Energy return in footwear–revisited,” 2024, *Taylor and Francis Ltd.* doi: 10.1080/19424280.2024.2369995.

[35] P. Fleming and S. Forrester, “Artificial turf research at Loughborough University,” in *Procedia Engineering*, Elsevier Ltd, 2014, pp. 925–930. doi: 10.1016/j.proeng.2014.06.147.

[36] “GreenFields USA W weave.” Accessed: Aug. 13, 2025. [Online]. Available: https://greenfieldsusa.com/innovation/

[37] H. McGowan, P. Fleming, J. Mcmahon, D. James, and S. Forrester, “The effect of normal stress on rotational traction on artificial turf surfaces,” Sep. 2024, doi: 10.17028/RD.LBORO.27052183.V1.

[38] “The next generation of synthetic turf: the future of non-fill systems - TenCate.” Accessed: Aug. 13, 2025. [Online]. Available: https://tencategrass.com/expertise/the-next-generation-of-synthetic-turf-tencate-grass-strongly-believes-in-the-future-of-non-fill-systems/

[39] “Non-Infill Turf Systems - GreenFields GreenFields Europe.” Accessed: Aug. 13, 2025. [Online]. Available: https://www.greenfields.eu/products/non-infill

[40] “Infill or Non-Infill Artificial Grass: Making the Right Choice,” *https://arturf.com/*, Accessed: Aug. 13, 2025. [Online]. Available: https://arturf.com/buyers-guide/infill-or-non-infill-artificial-grass/

[41] “4.1 SISTurf Xtreme Ultra - Data Sheet”.

[42] “FUTURE 7 ULTIMATE FG/AG Football Boots Unisex | PUMA.” Accessed: Aug. 16, 2025. [Online]. Available: https://uk.puma.com/uk/en/pd/future-7-ultimate-fg-ag-football-boots-unisex/107916

[43] “Puma | Future 7 Ultimate Firm Ground Football Boots | Firm Ground Football Boots | SportsDirect.com.” Accessed: Aug. 16, 2025. [Online]. Available: https://www.sportsdirect.com/puma-future-7-ultimate-firm-ground-football-boots-203351#colcode=20335125

[44] “Force plate: function and applications | Kistler GB.” Accessed: Aug. 17, 2025. [Online]. Available: https://www.kistler.com/GB/en/force-plate/C00000113

[45] “kistler”.

[46] "PDF downloads for Vicon Nexus - Nexus 2.16 documentation - Vicon Help." Accessed: Aug. 15, 2025. [Online]. Available: https://help.vicon.com/space/Nexus216/11611993/PDF+downloads+for+Vicon+Nexus?attachment=https://help.vicon.com/download/attachments/11611993/Vicon%2520Nexus%2520User%2520Guide.pdf&type=application/pdf&filename=Vicon+Nexus+User+Guide.pdf

[47] J. W. Wannop, E. S. Graf, and D. J. Stefanyshyn, "The effect of lateral banking on the kinematics and kinetics of the lower extremity during lateral cutting movements," *Hum Mov Sci*, vol. 33, no. 1, pp. 97–107, Feb. 2014, doi: 10.1016/J.HUMOV.2013.07.020.

[48] G. Luo and D. Stefanyshyn, "Identification of critical traction values for maximum athletic performance," *Footwear Sci*, vol. 3, no. 3, pp. 127–138, 2011, doi: 10.1080/19424280.2011.639807.

[49] C. Stratford, T. Dos'Santos, and J. J. McMahon, "The 10/5 Repeated Jumps Test: Are 10 Repetitions and Three Trials Necessary?," *Biomechanics (Switzerland)*, vol. 1, no. 1, pp. 1–14, Mar. 2021, doi: 10.3390/biomechanics1010001.

10 Appendix

Combined outcome measures that were attempted

Participant	Surface	CoT_plateau	Peak CoT (0–20%)	Peak Vertical RFD (kN/s)	Peak Horizontal RFD (kN/s)	Fz_peak (N)	Fz_peak / kg	Fz_peak / BW	Fh_peak (N)	Fh_peak / kg	Fh_peak / BW
1	SBR	0.627	0.917 ± 0.080	51.49 ± 30.42	33.94 ± 19.78	1993.07	24.37	2.48	1427.73	17.45	1.78
	PurePT Fresh	0.648	0.933 ± 0.073	45.02 ± 21.07	30.18 ± 12.91	2059.06	25.17	2.57	1434.86	17.54	1.79
	PurePT Worn	0.605	0.858 ± 0.097	62.80 ± 12.85	37.61 ± 6.65	2728.16	33.35	3.4	1725.51	21.09	2.15
2	SBR	0.838	0.999 ± 0.065	36.99 ± 9.84	30.37 ± 6.68	1507.3	19.91	2.03	1392.5	18.4	1.88
	PurePT Fresh	1	2.972 ± 0.671	28.70 ± 8.64	23.28 ± 5.50	1410.8	18.64	1.9	1389.7	18.36	1.87
	PurePT Worn	0.84	1.002 ± 0.044	21.05 ± 6.54	19.03 ± 5.53	1522.5	20.11	2.05	1415.9	18.7	1.91
3	SBR	0.792	0.932 ± 0.053	50.73 ± 10.70	34.43 ± 7.55	1800.71	22.01	2.139	1390.83	16.21	1.652
	PurePT Fresh	0.836	1.000 ± 0.061	52.99 ± 4.63	32.76 ± 2.93	1789.95	20.86	2.127	1435.35	16.73	1.705
	PurePT Worn	0.848	1.019 ± 0.036	56.68 ± 12.04	34.20 ± 4.91	1723.57	20.09	2.048	1478.79	17.24	1.757

Weight Acceptance Torque (N·m)	Midstance Torque (N·m)	Peak Push-off Torque (N·m)	Final Push-off Torque (N·m)	Rearfoot Deformation Max Mean (mm)	Rearfoot Max Trial (mm)	Forefoot Deformation Max Mean (mm)	Forefoot Max Trial (mm)	Yaw Change WA (°)	Yaw Midstance (°)	Yaw Push-off (°)	Yaw Final (°)	Forefoot AP Slip (mm)	Forefoot ML Slip (mm)	Rearfoot AP Slip (mm)	Rearfoot ML Slip (mm)
335.48	203.32	177.54	102.83	48.29	71.48	23.71	30.85	18.47	1.18	0.05	15.75	36.27	19.69	42.52	19.46
166.44	130.77	101.18	58.97	57.9	91.39	28.84	48.47	24.23	0.1	-0.3	21.94	82.75	68.32	98.92	0
309.08	207	152.42	64.65	62.24	75.71	30.83	38.86	37.14	-0.09	-1.93	24.34	21.02 (IC→TO)	8.41 (IC→TO)	35.50 (IC→TO)	73.59 (IC→TO)
83.04	101.97	113.84	76.11	51.78	75.05	23.94	30.44	43.74	2.99	-19.02	-2.38	12.82 (IC→TO)	4.72 (IC→TO)	-15.99 (IC→TO)	49.67 (IC→TO)
44.78	62.4	87.05	82.48	65.18	78.03	21.51	35.33	29.4	0.77	-13.6	-6.3	40.47 (IC→TO)	-0.41 (IC→TO)	62.50 (IC→TO)	18.41 (IC→TO)
382.99	279.84	165.95	25.36	—	—	—	—	22.53	0.22	-5.67	-11.95	30.30 (IC→TO)	-0.14 (IC→TO)	30.05 (IC→TO)	21.64 (IC→TO)
447.5	312.83	327.05	151.09	—	—	—	—	18.47	1.18	0.05	15.75	4.13 (IC→TO)	-19.05 (IC→TO)	-1.47 (IC→TO)	19.80 (IC→TO)
109.71	15.75	-40.01	6.31	—	—	—	—	24.23	0.1	-0.3	21.94	34.70 (IC→TO)	-4.53 (IC→TO)	32.75 (IC→TO)	29.66 (IC→TO)
—	—	—	—	—	—	—	—	37.14	-0.09	-1.93	24.34	40.00 (IC→TO)	-15.00 (IC→TO)	4.13 (IC→TO)	24.94 (IC→TO)